

Tracing the International Transmission of a Crisis through Multinational Firms

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ABSTRACT

We show that multinational firms transmit shocks across countries through their internal capital markets. We study a credit supply shock to parent firms in Germany. International affiliates outside Germany supported their parents through internal lending, became financially constrained themselves, and experienced lower real growth. We find that managers were “Darwinist” with respect to international affiliates but “Socialist” in the home country, that internal capital markets transmitted the credit shock more strongly than a nonfinancial shock, and that access to developed credit markets attenuated the real effects. The total real impact of shock transmission through multinationals on foreign economies was large.

THE INTERNAL CAPITAL MARKETS OF multinational firms crucially shape global capital flows. In recent years, internal capital flows between multinational parents and their international affiliates accounted for over 50% of total capital inflows into the median country and amounted to 4% of gross domestic product (GDP) in the median country.¹

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¹ Multinational parents and affiliates together produce 36% of global output, while foreign-owned affiliates alone produce 11% of global output and 8% of U.S. output (OECD Analytical Activity of Multinational Enterprises (AMNE) database for 2016; Broner et al. (2010)).

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Despite their size and ubiquity, it has not been fully established how internal capital markets of multinationals operate, whether they impact the real economy, and which frictions play a role. The key challenges are that data on internal capital flows are scarce and that causal identification is challenging. We provide new evidence by combining detailed microdata on internal capital positions with a quasi-experimental research design.

We study a lending cut by Commerzbank, a large German bank, that directly reduced the credit supply of parent corporations of multinationals located in Germany but not the credit supply of international affiliates of these multinationals. We show that German parents indirectly transmitted the credit shock to their international affiliates. Relative to unaffected affiliates, sales and employment of international affiliates with affected parents fell sharply and remained low for three years before subsequently recovering. The transmission of Commerzbank's lending cut through multinationals had large effects on countries outside Germany. For instance, the total impact amounted to roughly 0.4% of aggregate sales in Austria and the Czech Republic and 0.1% in Spain and the United Kingdom.

We present evidence that internal capital markets contributed to shock transmission, as the real effects were concentrated among affiliates that increased internal lending to their parent and became financially constrained themselves. Weak international affiliates were hit more strongly, suggesting that managers of multinationals are relatively "Darwinist" with respect to international affiliates. In contrast, affiliates within Germany were not significantly harmed, even if they were weak, implying that managers have home bias and relatively "Socialist" preferences toward home country affiliates. Frictions due to currency, geography, and capital controls were not important, but developed external credit markets helped affiliates to partially attenuate the negative real effects. In contrast to the lending cut, a nonfinancial flooding shock to parents had only weak effects on international affiliates, suggesting that internal capital markets transmit financial shocks, which harm access to external credit, more forcefully than nonfinancial shocks.

Identifying the role of multinationals and internal capital markets in shock transmission is difficult. The key empirical challenge is that parents and their affiliates are subject to common shocks. For example, parents and affiliates frequently use similar raw materials, sell similar products, and share similar productivity and other characteristics (Guadalupe, Kuzmina, and Thomas (2012), Fort et al. (2013), Foster, Grim, and Haltiwanger (2016)). Shocks to common input prices, to common product demand, and to firms with similar characteristics could induce comovement between parents and affiliates, even if shocks to the parent are not transmitted to affiliates.

We overcome the empirical challenge by studying a shock that directly affected only parents of German multinationals and not their international affiliates. The shock was a lending cut by Commerzbank, a large German bank. Commerzbank experienced significant losses on its financial investments during the 2008/09 financial crisis. The losses were caused by a combination of misjudgments and bad luck in the international investment

and trading divisions. Crucially, the origin of the losses was independent of Commerzbank's corporate lending division. We present a range of evidence that firms borrowing from Commerzbank would have been on similar growth paths as borrowers of other German banks had the lending cut not happened. The lending cut eventually affected corporate borrowers, however, because the losses forced Commerzbank to reduce loan supply to its borrowers.

Commerzbank's exposure to the crisis and its lending cut were severe compared to other German banks.² Existing evidence shows that European firms of all sizes, including multinationals, struggle to raise financing when a relationship bank cuts lending (Bentolila, Jansen, and Jiménez (2018), Huber (2018)). As a result, Commerzbank's lending cut was an exogenous shock to the credit supply of German firms with a preexisting relationship to Commerzbank, relative to German firms borrowing from other banks. Parents with higher pre-crisis dependence on Commerzbank reduced bank debt, did not use alternative sources of external financing, and experienced lower sales from 2008 to 2010. In contrast, the credit supply of international affiliates of these multinationals was not directly affected, since Commerzbank's corporate lending was concentrated in Germany.

We investigate whether Commerzbank's lending cut was indirectly transmitted abroad by German multinationals. We study international affiliates whose German parents were directly hit by the lending cut. Our estimation sample exclusively contains affiliates of German parents in other countries. We test whether affiliates whose German parents were more dependent on Commerzbank grew differently relative to other affiliates. We always compare affiliates located in the same country at the same point in time, so that differences in demand or other country-specific shocks do not affect the estimates.

In the first set of affiliate-level results, we analyze measures of affiliate growth, starting with sales. Capital constraints may lower sales in the short run because firms have to cut variable inputs, cannot fund the distribution of output to points of sale, or focus on rebuilding liquidity buffers (Calomiris, Himmelberg, and Wachtel (1995)). We find that sales of affiliates with greater parent Commerzbank dependence dropped sharply once Commerzbank reduced lending in 2008 and took until 2011 to fully recover. Similarly, employment and holdings of short-term, production-related assets (e.g., raw materials, working capital) were lower from 2008 to 2010 and recovered after 2011. There were no significant effects on external financing (e.g., bank debt or external equity), consistent with the view that financial frictions make it difficult for affiliates to raise funding in the short run (Catherine et al. (2022)). Taken together, the results show that real outcomes of international affiliates, without a direct borrowing relationship to Commerzbank, were harmed by the shock to Commerzbank.

² The vast majority of German banks had not invested in international financial markets. A few other banks also suffered losses in 2008/09 (Puri, Rocholl, and Steffen (2011)). For idiosyncratic reasons, these other banks are not well suited to identifying a quasi-experimental shock to firm credit supply, as we discuss in Section I and Internet Appendix III. The Internet Appendix may be found in the online version of this article.

Several findings suggest that affiliates with greater parent Commerzbank dependence would have evolved in parallel to other affiliates had Commerzbank's lending cut in Germany not happened. We find no effect for affiliates with positive but low parent Commerzbank dependence because their parents could borrow from other relationship banks. This result shows that parent Commerzbank dependence was not generally associated with slow growth. The results are also robust to controlling for country-industry-time fixed effects as well as other characteristics of affiliates and parents (e.g., size, financial ratios, and number of banks) interacted with time. In addition, affiliates with greater parent Commerzbank dependence were not different on observable characteristics, were not on different growth paths before the lending cut, did not borrow directly from Commerzbank, and did not face a credit shock themselves.

In the second set of affiliate-level results, we investigate the response of internal capital flows through multinationals. Data from Deutsche Bundesbank report detailed internal capital market positions between each international affiliate and their German parent. There is significant heterogeneity in how much affiliates financially interact with the parent. Some affiliates raise funding in their host countries and lend to the parent, while others do not lend internally. We begin by considering affiliates that had issued internal long-term loans to their parent before 2008. Parents had already set up a financial infrastructure to borrow from these affiliates, making it possible to quickly shift capital from these affiliates toward affected parents after 2008. We show that affiliates with previous internal loans strongly increased lending to their parent after the lending cut, but other affiliates did not. Investigating real effects, we find that the reduction in affiliate sales was large and significant for affiliates that had previous internal loans and increased internal lending, but insignificant for other affiliates. This pattern is not driven by other affiliate and parent characteristics. Overall, the results suggest that increases in internal lending played a role in transmitting the lending cut to affiliates.

We also study whether declines in internal trade contributed to decreases in affiliate sales. We construct a proxy for affiliates that traded internally by selling goods to the parent. We find evidence that lower internal sales from affiliates to parent contributed to the sales decline at affected affiliates. At the same time, we show that internal trade does not explain the entire effect. For instance, we control for internal trade credit to the parent and analyze only affiliates that were unlikely to trade with the parent (e.g., affiliates in the same industry as the parent, in the service sector, and not in wholesale and retail industries). In all these specifications, affiliates with greater internal lending suffered large and significant sales declines, while the effects on other affiliates were relatively small and insignificant. Taken together, the results suggest that both internal capital flows and internal trade transmitted the shock to affiliates.³

³ Since we analyze multiple outcomes, we show that the results are not driven by a multiple inference problem, using the *p*-value adjustment by Anderson (2008). All adjusted *p*-values remain below 0.1 if they were below 0.1 without adjustment.

In the third set of affiliate-level results, we examine mechanisms and frictions that shape the response and real effect of internal capital markets. In a standard model, such as the model we present in [Internet Appendix II](#), managers of parents treat international affiliates in a “Darwinist” manner and favor affiliates with the greatest growth potential (Stein (1997), Inderst and Mueller (2003)). Consistent with this model, we find that international affiliates with higher growth before the lending cut were significantly less affected by the lending cut to their parent. However, managers seemed to follow a different model within Germany. The negative effects on affiliates located within Germany were relatively weak on average, suggesting that parent managers have home bias. In addition, effects on affiliates within Germany were not associated with growth potential, implying a managerial preference for “Socialism” in the home country (Matvos and Seru (2014)). These findings are consistent with models in which managers are exposed to power struggles and rent-seeking by affiliates in their home country (Rajan, Servaes, and Zingales (2000), Scharfstein and Stein (2000)) but in which these forces are weaker in the case of international affiliates.

We show that credit market frictions shaped the real effects on affiliates, as the effects were weaker in countries with developed credit markets (high credit-GDP ratio). However, we still find a significant decline in affiliate sales in high-credit countries, consistent with the view that affiliates are also borrowing-constrained in developed markets (Lian and Ma (2020)). Frictions due to different currencies, geographic distance between parent and affiliate, or local business cycles did not significantly affect the transmission to affiliates, suggesting that multinationals are able to shift capital relatively freely across the world. Moreover, regulation on capital flows did not play an important role, likely because regulation typically does not strictly limit internal capital flows.

In the fourth set of affiliate-level results, we consider how internal capital markets respond to nonfinancial shocks. Firms exposed to financial shocks cannot rely on their own external credit supply to overcome their financial constraints, forcing them to transmit the shocks to affiliates. In contrast, the effects of nonfinancial shocks may differ because directly shocked firms can access external funding without harming affiliates. To examine this possibility, we identify parents that were hit by a large-scale flood in 2013 but whose relationship banks were still able to lend to them (Koetter, Noth, and Rehbein (2020)). We find that the flood harmed flooded parents’ total assets, but that these parents substantially increased their bank debt, implying that they were not financially constrained. As a result, parents did not need to withdraw capital from international affiliates and affiliate sales did not significantly fall after the flood. Taken together, the evidence underscores the importance of access to finance: internal capital markets transmit shocks forcefully when directly shocked firms cannot access external credit supply, but less strongly when external funding is available to smooth the shock.

In the final part of our analysis, we discuss in which countries the transmission of Commerzbank’s lending cut through German multinationals had the largest real impact. Specifically, we calculate how much greater total sales of

German affiliates in a given country would have been if affiliates with positive parent Commerzbank dependence had grown at the same rate as affiliates with zero parent Commerzbank dependence. This measure of “impact” amounted to 0.5% of aggregate sales in the Czech Republic, 0.4% in Austria, and 0.3% in Poland.⁴ These numbers highlight that a shock to an individual firm in one country (Commerzbank in Germany) can have first-order effects on the distribution of firm growth in many other countries, solely because of transmission through the internal networks of multinationals.

The remainder of the paper describes related literature, the shock to Commerzbank, the data, the empirical strategy, the effects on affiliate growth, the importance of internal capital markets, mechanisms and frictions driving internal capital markets, a comparison to a nonfinancial flooding shock, and the impacts across different countries.

Related Literature

A large literature analyzes capital allocation within domestic business groups (Lamont (1997), Shin and Stulz (1998), Gopalan, Nanda, and Seru (2007), Boutin et al. (2013), Matvos and Seru (2014), Seru (2014), Almeida, Kim, and Kim (2015), Giroud and Mueller (2015, 2019), Santioni, Schiantarelli, and Strahan (2020), Giroud et al. (2021)). A literature on multinationals shows that affiliates of multinationals outperform domestically owned firms during domestic crises and when domestic capital markets are weak (Desai, Foley, and Hines (2004), Desai, Foley, and Forbes (2008), Foley and Manova (2015), Manova, Wei, and Zhang (2015), Kalemli-Özcan, Kamil, and Villegas-Sanchez (2016)). Recently, Boehm, Flaaen, and Pandalai-Nayar (2019) and Bena, Dinc, and Erel (2022) find that multinationals transmit shocks across countries, but do not focus on the role of internal capital markets and financial shocks.

Our paper makes several contributions relative to existing work. First, we present evidence that internal capital market flows on their own can transmit shocks across affiliates. Most previous work is consistent with the view that other transmission channels or differences across affiliates are responsible for shock transmission across firms and countries, instead of internal capital markets.⁵ We are able to take a step toward isolating the role of internal capital markets because we combine direct data on which affiliates lend internally with a quasi-experiment. Second, we analyze how internal capital flows across

⁴ The impact numbers reported here may not equal aggregate changes because of general equilibrium effects, as we detail below. However, recent research suggests that aggregate changes may be of the same order of magnitude or larger than the negative impacts reported by us (Chodorow-Reich (2014), Huber (2018), Herreño (2020), Sraer and Thesmar (2023)).

⁵ For example, internal trade and the allocation of management, human capital, machinery, and other resources may explain the transmission results in Shin and Stulz (1998), Giroud and Mueller (2019), and Bena, Dinc, and Erel (2022). Differences in productivity, human capital of workers, and reliance on domestic inputs may explain why affiliates of multinationals outperform other firms (Bloom, Sadun, and Van Reenen (2012), Guadalupe, Kuzmina, and Thomas (2012), Setzler and Tintelnot (2021)).

countries depend on different frictions than those across domestic business groups (e.g., access to external finance, “Darwinism” internationally vs. “Socialism” at home, currencies, geography, and capital controls.)

Third, we highlight that internal capital markets transmit financial shocks strongly and that recovery takes several years. This suggests that internal capital markets are a quantitatively important channel through which financial shocks propagate across countries. While much of the literature shows that internal capital markets reduce macroeconomic volatility in affiliate host countries, we show that internal capital flows can at times raise volatility and lower growth.

The paper also contributes to our understanding of how individual firms affect the international economy. The literature suggests that “granular” firms affect aggregate growth in their home country, as studied theoretically (Gabaix (2011), Acemoglu et al. (2012), di Giovanni and Levchenko (2012), Grassi (2017), Magerman et al. (2017), Carvalho and Grassi (2019)) and empirically (di Giovanni, Levchenko, and Mejean (2014), Friberg and Sanctuary (2016), Amiti and Weinstein (2018)). We highlight a new mechanism: a shock to an individual firm (in our case, Commerzbank) can impact firms all over the world, even if the shocked firm initially transmits the shock only within its home country. This mechanism is novel because it generates international shock transmission without requiring that a firm be directly connected to other countries.⁶ Our methodology is distinct from existing work because we quantify how important an individual firm can be, rather than the joint impact of all firm-level shocks.

Finally, the results of this paper suggest that comovement of connected countries (Frankel and Rose (1998)) and firms (Budd, Konings, and Slaughter (2005), Desai and Foley (2006)) partially reflects causal shock transmission rather than just common shocks across countries (Imbs (2004)). The results also support models in which multinationals increase cross-country comovement (Contessi (2010), Zlate (2016), Menno (2017), Tintelnot (2017)).⁷

I. The Shock to Commerzbank

This paper investigates how a shock to parents was transmitted internationally. In this section, we describe the specific shock that we analyze: the lending

⁶ The mechanism is distinct from the finding that firms comove with foreign countries where they have direct trade or production links (di Giovanni, Levchenko, and Mejean (2018, 2024), Hassan et al. (2024)). The mechanism also differs from the result that multinational banks with direct operations in several countries transmit crises internationally (Peek and Rosengren (1997), Acharya and Schnabl (2010), Cetorelli and Goldberg (2012), Popov and Udell (2012), Schnabl (2012), de Haas and van Lelyveld (2014), Ongena, Peydró, and van Horen (2015), di Giovanni et al. (2021)). Hassan et al. (2024) similarly find that risk is transmitted through channels not captured by direct trade links.

⁷ In theory, financial connections can lower comovement (Baxter and Crucini (1995), Kalemli-Özcan, Papaioannou, and Peydro (2013), Monnet and Puy (2019)) or raise comovement (Morgan, Rime, and Strahan (2004), Imbs (2006)), depending on whether credit demand or supply shocks are the primary source of fluctuations.

cut of Commerzbank, the second-largest German bank before 2008. Commerzbank's business model focused on corporate lending in Germany, with 96% of its branches located in Germany. Commerzbank's market share in lending to medium-sized (*Mittelstand*) and large firms in Germany was around 13%.⁸

The bank also ran trading and investment divisions that had heavily invested in international financial markets prior to the 2008/09 financial crisis. At the onset of the crisis, Commerzbank held large positions in U.S. mortgage-related securities (mortgage-backed securities and collateralized debt obligations) as well as institutional debt that would end up failing (including exposure to Lehman Brothers and the Icelandic banks). After U.S. mortgage markets crashed, these positions led to large losses in Commerzbank's trading and investment divisions.

Overall, Commerzbank's equity capital fell by a total of 68% between December 2007 and December 2009. The drop in equity was entirely due to two components: write-downs on financial instruments and income losses (Figure 1, Panel A). Write-downs on financial instruments were caused by changes in the valuation of derivatives held by the bank, and can therefore be attributed to the trading and investment divisions. The income losses were also driven by the trading and investment divisions (Figure 1, Panel B). Interest income, in contrast, which includes Commerzbank's earnings from lending to firms, remained on an upward trend up to 2009. These figures suggest that Commerzbank's losses were not caused by the corporate lending portfolio. Instead, Commerzbank's corporate borrowers were growing steadily right until Commerzbank cut lending.

Following its trading and investment losses, Commerzbank became financially constrained. It struggled to get wholesale funding in 2008 and was close to its equity threshold. As a result, it cut lending to German borrowers. The closest comparison group to Commerzbank is composed of other German commercial banks. They also operated for profit and had a comparable business model and similar types of borrowers. Commerzbank's lending to German borrowers grew in parallel to other commercial banks and all other German banks until 2007, but fell sharply thereafter (Figure 2). In contrast, lending by other German banks actually increased slightly from 2007 to 2009. There was no domestic financial crisis that affected all German banks, for example, there was no housing boom and bust. The majority of German banks suffered less from the 2008/09 financial crisis and were therefore able to continue lending (see [Internet Appendix III](#) for a detailed analysis of other banks).

⁸ None of the information on individual banks is provided by Deutsche Bundesbank. The main data sources are annual reports of Commerzbank and Dresdner Bank and financial analyst reports as listed at the end of [Internet Appendix I](#). Throughout the paper, "Commerzbank" refers to all branches that were part of the Commerzbank network in 2009, including its acquisition Dresdner Bank. Commerzbank had already decided to acquire Dresdner Bank before the crisis hit both banks severely. Both banks suffered significant losses in 2008. The lending cut affected firms that had banked with the old Dresdner Bank to a similar degree as firms that had banked with the old Commerzbank, so we construct the treatment variable based on relationships to either bank. See [Internet Appendix III](#) for more details.

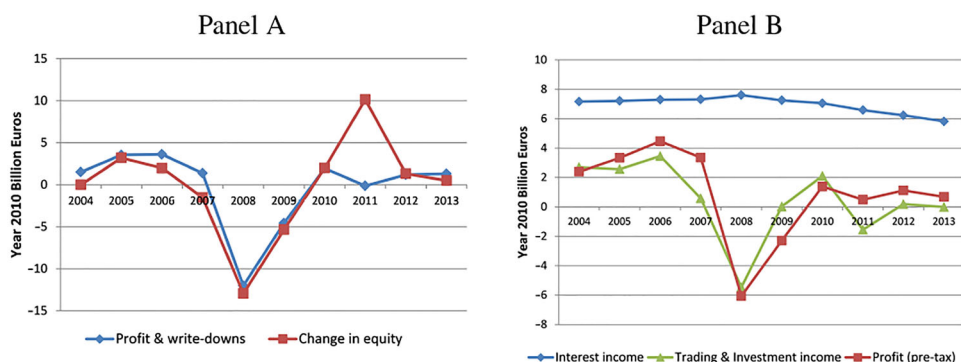


Figure 1. Commerzbank's equity capital, write-downs, and profits. The left panel shows Commerzbank's profits and write-downs plus equity capital. Write-downs arise from changes in revaluation reserves, cash flow hedges, and currency reserves. Panel B shows the composition of Commerzbank's profits. Interest income is interest received from loans and securities minus interest paid on deposits. Trading and investment income is the sum of net trading income, net income on hedge accounting, and net investment income. Pretax profit is interest income plus trading and investment income minus costs. The values are in 2010 billion euro. The positions of Commerzbank and Dresdner Bank for the years before the 2009 takeover are aggregated. Data Source: Bank Annual Reports, Huber (2018), own calculations. (Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jofi.13338))

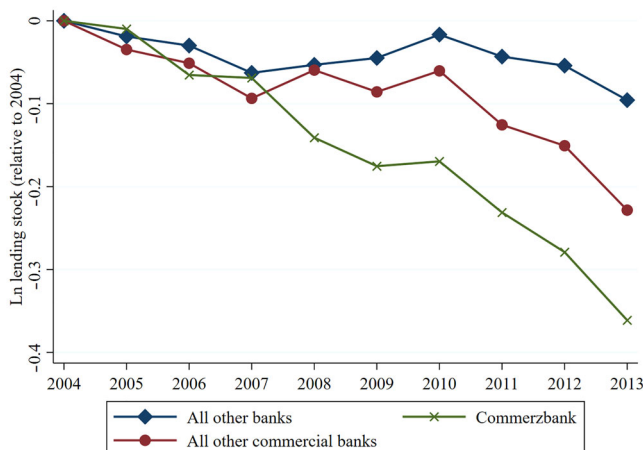


Figure 2. Lending by German banks. The figure plots the log lending stock to German non-financial customers for different groups of banks, relative to 2004. The values are in 2010 billion euro. We include lending by branches of Commerzbank and Dresdner Bank to calculate lending by Commerzbank (provided in the bank annual reports). We subtract lending by Commerzbank from aggregate lending by all German banks (provided by the Deutsche Bundesbank) to calculate lending by all other banks. We subtract lending by Commerzbank, savings banks, Landesbanken, and cooperative banks from aggregate lending by all German banks to calculate lending by all other commercial banks. Data Source: Bank Annual Reports, Deutsche Bundesbank, Huber (2018), own calculations. (Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jofi.13338))

Research reports by financial analysts support the view that Commerzbank's lending cut was an exogenous shock to German borrowers. These reports are useful because analysts communicate directly with bank managers and have access to information beyond banks' official reports. We study 110 analyst research reports extracted from the Thomson Reuters Investtext Database, listed in the final table of the [Internet Appendix](#). We extract all reports mentioning Commerzbank in 2008 and 2009, as well as relevant reports from years before and after. We systematically read the reports and use them to answer six questions about Commerzbank's lending cut. Our exact methodology and the results are described in [Internet Appendix Table IA.I](#).

There are six key lessons from the analyst reports. First, Commerzbank's loan portfolio did not contribute to the losses. In contrast, the performance of corporate borrowers was generally "strong" and generated "healthy returns" until Commerzbank's lending cut (e.g., 2008 reports by ESN/equinet and Deutsche Equity Research). Second, Commerzbank's borrowers were not any riskier or more cyclical than other banks' borrowers (JPMorgan (2007) and Natixis (2006)). Third, the investment and trading divisions operated separately from the loan division. There was no cross-hedging of risks and no common strategy, so the divisions were "conceptually separate" (CA Cheuvreux (2008)). The evolution of income from the different divisions also suggests that their performance was not correlated (Figure 1, Panel B).

The fourth lesson is that Commerzbank's managers recognized the severity of the U.S. mortgage crisis too late, so exposure to U.S. mortgage-related securities contributed to the losses (Credit Suisse Europe (2008)). Fifth, the bank's managers believed that governments would prevent large-scale institutional failures of Commerzbank's interbank partners, such as Lehman and the Icelandic banks, so they suffered severe losses when these banks failed (ESN (2009)). In combination, these exposures depleted Commerzbank's equity and forced it to cut lending. Commerzbank received equity injections by the government, but they were not quick and large enough to prevent a lending cut altogether. The sixth lesson is that it took Commerzbank until 2010 to fully stabilize. Its lending grew roughly in parallel with other commercial banks from 2010 onward.

II. Data

We trace out the international effects of Commerzbank's lending cut using three data sets: information on the relationship banks of German parents, balance sheets of international affiliates from the Microdatabase Direct Investment (MiDi, Deutsche Bundesbank (2017a)), and balance sheets of German parents from the Ustan database (Deutsche Bundesbank (2017b)).

A. Relationship Banks of German Parents

German corporate finance traditionally relies on a system of relationship banking. This means that German borrowers form close relationships with

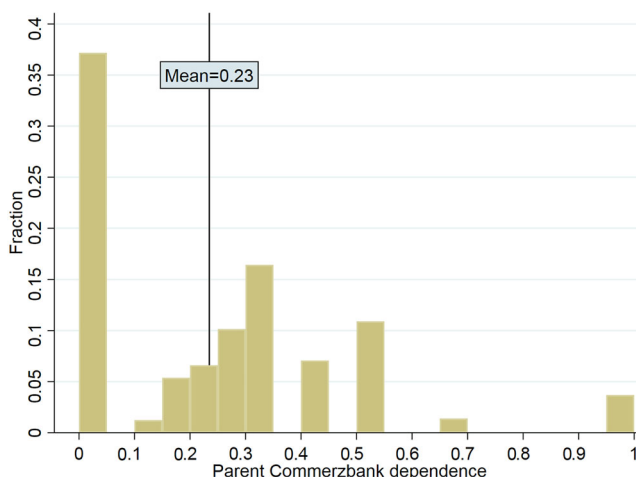


Figure 3. Distribution of parent Commerzbank dependence. The figure shows a histogram of Commerzbank dependence for the 655 German parents in our data set in 2006. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations. (Color figure can be viewed at wileyonlinelibrary.com)

a few relationship banks. In German, relationship banks are called “home banks” (*Hausbanken*), which highlights the tight ties between borrowers and banks. Bank-firm relationships are long-lasting, as each year only 1.7% of firms add a new relationship bank (Dwenger, Fossen, and Simmler (2015)). The most common services provided by relationship banks are loans and payment transactions (Elsas (2005)).

We obtain proprietary data on the names of the relationship banks (*Hausbanken*) of 112,344 German firms in 2006. The data are from the credit rating agency Creditreform, which collects information from firm surveys and financial statements. Our main treatment variable measures the fraction of a parent’s relationship banks that were Commerzbank branches:

$$Parent\ CB\ dep = \frac{\text{Number of parent's relationship banks that were CB branches}}{\text{Total number of parent's relationship banks}}. \quad (1)$$

Results are similar when we use the fraction of bank companies or an indicator for any Commerzbank relationship as the treatment variable. We do not have data on loan quantities from Commerzbank to individual parents. However, in the German system of relationship banking, information on relationship banks accurately predicts which parents were dependent on which banks.

The variable *Parent CB dep* is highly correlated with Commerzbank’s branching expansions after World War II and after German reunification, which suggests that it captures Commerzbank dependence accurately (Huber (2018)). Mean *Parent CB dep* was 0.23 and about 40% of parents had zero Commerzbank dependence (Figure 3).⁹

B. Data on Affiliates from MiDi

The data underlying MiDi are collected by Deutsche Bundesbank as part of its supervisory duties. German parents have to report an international affiliate to Deutsche Bundesbank if they hold at least 10% of an affiliate's equity and if the affiliate's total assets exceed 3 million euro. These reporting criteria have been constant since 2002, so we use data from 2002 until 2015, the most recent year at the time of data provision.

A unique feature of MiDi is that it contains detailed balance sheet positions on internal capital market positions between affiliates and parents. We can see annual measures of long-term loans from affiliate to parent, equity invested by the parent, total liabilities owed to the parent by the affiliate, and short-term claims by the affiliate on the parent. MiDi also includes annual data on affiliate balance sheets, sales, employment, and industry.¹⁰

Our estimation sample includes all affiliates that were directly owned by a German parent in 2006, the final year before the U.S. housing market crisis. We remove affiliates in the financial sector from the sample. Using a unique firm identifier, we match parents' relationship banks to MiDi (Schild, Schultz, and Wieser (2017)). We match the relationship banks for 26% of parents in MiDi. The other parents do not appear in the data on relationship banks. The affiliates of matched parents were responsible for 71% of total sales by international affiliates in MiDi in 2006. Overall, there are 655 German parents and 2,695 international affiliates in our data. The foreign direct investment of German parents was the third largest in the world in 2006 and the data contain affiliates located all over the world.

C. Data on Parents from Ustan

Ustan is a data set of annual balance sheets of nonfinancial German parents. The data are collected by Deutsche Bundesbank as a byproduct of its lending activity (Becker et al. (2019)). Ustan is also available from 2002 to 2015. Some parents that appear in MiDi do not appear in Ustan, so we supplement Ustan with data from Bureau van Dijk Orbis Historical Financials for these parents. Overall, we have balance sheet data on 407 of the 655 German parents in MiDi. The average parent had 3.8 international affiliates in 2006. There was no significant association between parent Commerzbank dependence and parent characteristics, conditional on industry (Table IA.II). Parents with positive Commerzbank dependence were on average larger, which arises mechanically because larger parents have more relationship banks (Table IA.III).

⁹ We cannot match data on the bank relationships of international affiliates to the MiDi data. However, we show using several tests that affiliates were not directly exposed to Commerzbank's lending cut in Section VI.F.

¹⁰ We winsorize all outcome variables at the first and 99th percentiles of their distribution to mitigate the impact of outliers. However, we find similar coefficients without winsorizing. Since some outcome variables are in logs and contain a handful of zeros, we add 1 unit (1,000 euro) to all log outcome variables throughout the paper. Tables IA.VII and IA.XXV show that the results are similar using the inverse hyperbolic sine transformation.

Table I
Affiliate Summary Statistics by Bins of Parent Commerzbank Dependence

The table shows means (standard deviations) for affiliates. Sales and Total Assets are in thousand euros. Leverage is defined as liabilities divided by total assets. The balance sheet items are in percent of total assets. The number of affiliates in the bottom row refers to the number of affiliates in MiDi in 2006. All values are for 2006. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations.

	Range of Parent Commerzbank Dependence				
	0	0.01–0.25	0.26–0.50	0.51–1	Total
Sales	42,184 (107,225)	50,056 (131,195)	65,406 (178,667)	52,495 (128,354)	54,400 (147,351)
Employment	162 (434)	222 (530)	200 (518)	211 (452)	196 (496)
Total assets	69,818 (280,584)	75,853 (303,662)	118,348 (426,055)	97,095 (353,319)	93,160 (357,546)
Leverage (%)	52.10 (33.75)	50.70 (32.20)	52.81 (34.60)	49.18 (32.74)	51.83 (33.65)
Long-term loans to parent (%)	0.38 (3.28)	0.15 (2.19)	0.44 (3.53)	1.92 (7.04)	0.46 (3.60)
Equity from parent (%)	14.11 (22.82)	14.09 (19.27)	15.36 (22.44)	15.13 (21.33)	14.69 (21.71)
Liabilities toward parent (%)	11.24 (21.01)	13.76 (22.27)	13.38 (22.25)	8.37 (16.56)	12.54 (21.60)
Short-term claims on parent (%)	4.40 (12.51)	4.81 (12.80)	3.81 (11.17)	3.45 (10.40)	4.19 (11.91)
Ownership share of parent in affiliate	0.878 (0.240)	0.870 (0.234)	0.884 (0.237)	0.842 (0.276)	0.876 (0.240)
Number of affiliates	721	675	1,101	198	2,695

D. Financial Statistics of Affiliates

The average affiliate was medium-sized, with annual sales of 54 million euro and 196 employees (Table I). It raised substantial financing on its own. For example, it borrowed from banks in its host country, as 76% of liabilities were owed to external, nonparent lenders. Some affiliates lent funds to their parents, while others did not lend internally.¹¹ On average, parents owned 88% of an affiliate's equity. They therefore had the ability to influence affiliates' financial decisions and steer internal lending.

There was no significant association between parent Commerzbank dependence and affiliate sales, employment, leverage, internal capital positions, and parent ownership share, conditional on affiliate country and industry (Table IA.IV). The only statistically significant coefficient is on total assets, but the point estimate is small, implying that a 1% increase in assets is associated

¹¹ We find that internal lending depends on external borrowing costs in the affiliate host country (difference between lending and deposit rate). Specifically, net lending falls with the spread, whereas funding by the parent rises with the spread (Table IA.VI).

with a 0.00013 increase in parent Commerzbank dependence. Affiliates of parents with positive Commerzbank dependence also operated in similar industries and countries as other affiliates (Table IA.V).¹²

III. Empirical Strategy

In this section, we outline how we trace the effects of Commerzbank's lending cut across countries. In short, we identify effects on multinational parents borrowing from Commerzbank in Germany and then investigate the effects on international affiliates.

A. Empirical Challenge

The empirical challenge in identifying the transmission of shocks is that there may be unobserved shocks that simultaneously hit parents and their affiliates. Consider a global decrease in demand for a particular car model produced by a German multinational. The parent that produces the less popular model in Germany would produce and sell fewer cars in Germany. At the same time, affiliates that produce the same model in other countries would produce and sell fewer cars in their host countries. Even without a causal transmission channel from parent to affiliates, parent and affiliate growth would comove.

A related issue arises from the fact that size and productivity of parents and affiliates are positively correlated (Guadalupe, Kuzmina, and Thomas (2012)), while firm outcomes also differ systematically by size and productivity (Fort et al. (2013), Foster, Grim, and Haltiwanger (2016)). This again suggests that parents and affiliates are often exposed to common shocks and that comovement does not imply shock transmission.

Common shocks could also lead to the spurious concurrence of internal capital market flows with shocks to the parent. For instance, if both affiliate and parent experience a shock to the demand for their products, the affiliate may transfer some of the extra revenue to its parent. As a result, an internal capital flow would appear simultaneously with a shock to the parent, although the internal capital flow was not triggered by the shock to the parent and did not contribute to shock transmission from parent to affiliate.

B. Commerzbank's Lending Cut and Relationship Banking

To overcome the challenge of common shocks, we need to identify a shock that affected parents without directly impacting international affiliates. Commerzbank's lending cut during the 2008/09 financial crisis provides such a shock.

¹² We do not report data by bins of Commerzbank dependence, as above, because disclosure rules of Deutsche Bundesbank do not allow us to report statistics for cells that contain only a few firms.

A subset of German parents had Commerzbank as relationship bank. A large theoretical literature shows that relationship banks gain an informational advantage from their long-standing dealings with relationship borrowers (Sharpe (1990), Boot (2000)). This reduces asymmetric information and improves banks' monitoring capabilities. But the informational advantage creates an adverse selection problem when borrowers want to find new lenders. New potential lenders suspect that firms not getting loans from their existing relationship bank are of low quality. This problem is particularly severe in a recession like the 2008/09 crisis when default rates are high (see also Slovin, Sushka, and Polonchek (1993), Lemmon and Roberts (2010), and Alfaro, García-Santana, and Moral-Benito (2021)). At the time, it was not generally known that Commerzbank's losses were caused by external factors and not the performance of German borrowers. As a result, Commerzbank's borrowers struggled to get credit when their bank cut lending.

Commerzbank's corporate lending was concentrated on borrowers in Germany. The bank credit supply of international affiliates of German multinationals was therefore not directly harmed by Commerzbank's lending cut.¹³ In our main analyses, we compare the growth of affiliates, whose German parents were relatively dependent on Commerzbank for financial services, to the growth of other affiliates, whose German parents had lower or zero Commerzbank dependence.¹⁴ Our estimation sample exclusively contains affiliates of German multinationals located outside Germany. We always compare affiliates located in the same country at the same time.

C. Commerzbank's Lending Cut Affected German Parents

Our empirical strategy requires that Commerzbank's lending cut reduced the credit supply of German parents for whom Commerzbank was an important relationship bank. Existing evidence suggests that multinational parents in Europe depend on the credit supply of their relationship banks, for instance, Bentolila, Jansen, and Jiménez (2018) for Spain, Huber (2018), Berg, Reisinger, and Streit (2021), and Fraccaroli and Pizzigolotto (2022) for the case of Commerzbank, and general media reports on German multinationals (e.g., Reuters (2008, 2018)). While these papers already support our empirical approach and while the Ustan sample is somewhat small, it is still useful to examine the 406 parents in Ustan. (The MiDi affiliate data set used in the main analysis below is larger.)

¹³ To support this point, we show in Table III that affiliates with greater parent Commerzbank dependence did not reduce borrowing from banks after Commerzbank's lending cut. Moreover, the results are robust to excluding countries in which Commerzbank had an international office or business dealings with local banks (see Table IA.XX).

¹⁴ We estimate the reduced-form impact of the lending cut on affiliates and do not estimate the structural relationship between bank loans and borrower outcomes. The reason is that a lending cut can affect borrowers through many channels, including loan amounts, interest rates, the length of loan commitments, and uncertainty about future loan supply. Commerzbank dependence is a valid proxy for the overall effect of exposure to a lending cut, but not a valid instrument for any individual channel.

We find that the total bank debt of parents with greater Commerzbank dependence fell after Commerzbank's lending cut. The estimate is significant at the 10% level and implies that bank debt of the average parent with a Commerzbank relationship was 34.7 log points (roughly 29%) lower after the lending cut (Table IA.VII, column (1)).¹⁵ Before the lending cut, there was no association between parent Commerzbank dependence and bank debt growth, including during the 2003 recession (Internet Appendix Figure IA.1). This shows that parents with higher Commerzbank dependence were on parallel growth paths to other parents and not more cyclical. Similarly, firms with higher Commerzbank dependence reported more restrictive bank loan supply during Commerzbank's lending cut in a survey by the Ifo institute, but not during previous years (Table IA.VIII).

Between 2008 and 2010, parents were unable to compensate for the lost bank debt using nonbank financing (Table IA.IX, columns (1) to (3)). After 2011, external trade credit increased (Figure IA.2, significant at 1%) and compensated for around 79% of the lost bank debt, in line with the view that it takes time to raise external credit (Antràs and Foley (2015), Paravisini et al. (2015)). Consistent with this timing, parent sales were lower from 2008 to 2010 (significant at 10%), but recovered after 2011 (Table IA.IX, column (4)). Taken together, the results suggest that parents became financially constrained and it took them until roughly 2011 to overcome the constraints.

In the remaining analysis of this paper, we take as the starting point the observation that Commerzbank's lending cut caused an exogenous financial shock to German parents.

D. Support for the Identification Assumption

Our main empirical analysis tests whether affiliates with high parent Commerzbank dependence grew more slowly and increased internal lending after Commerzbank's lending cut. The main treatment variable is the Commerzbank dependence of affiliates' German parent in 2006. Our analysis identifies the causal effect of parents' exposure to the lending cut under a parallel-trends assumption: affiliates whose parents had high Commerzbank dependence would have evolved in parallel to other affiliates had the parents not been exposed to Commerzbank's lending cut.

Several factors suggest that this assumption holds. First, Commerzbank's lending cut was exogenous to German parents and their international

¹⁵ Figure IA.1 implies that the effects on parents' bank debt persisted until 2015, even though Commerzbank recovered by 2011 (Section I). An explanation for the persistence could be that parents voluntarily reduced their bank debt after the lending cut because managers were "scarred" by the experience of the lending cut. This is consistent with evidence that firms use less external financing when their managers personally experienced a credit crisis (Graham and Narasimhan (2004), Malmendier, Tate, and Yan (2011)) and with reports that German firms preferred alternative means of financing after 2009 because bank loan supply had proven to be risky (Fuchsbriefer (2018)). The bank debt point estimate is also relatively large, although at the 90% level we cannot reject the possibility that the effect of parent bank debt was just as large as Commerzbank's lending cut.

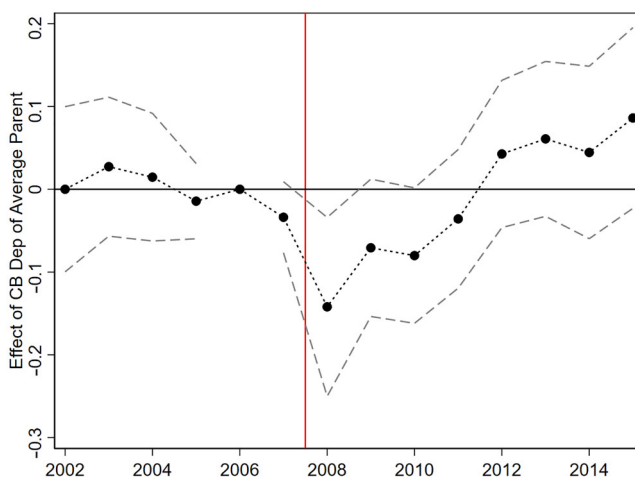


Figure 4. Impact of parent Commerzbank dependence on affiliate sales. The figure plots coefficients on *Parent Commerzbank Dependence*, interacted with year fixed effects. The gray, dashed lines represent 90% confidence intervals. The coefficients are estimated in a single regression. The outcome is $\log(\text{Affiliate sales})$. The following time-invariant control variables are calculated for affiliates in 2006 and interacted with a full set of year fixed effects: industry fixed effects, fixed effects for deciles of total sales, fixed effects for deciles of leverage, and country fixed effects. The specification also contains affiliate and year fixed effects. The coefficients are scaled to reflect the effect on an affiliate whose parent had average Commerzbank dependence, which was 0.23. Standard errors are two-way clustered at the level of the country and the parent. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations. (Color figure can be viewed at wileyonlinelibrary.com)

affiliates, as detailed in Section I. Second, affiliates with greater parent Commerzbank dependence were on parallel growth paths right up until Commerzbank's lending cut (Figure 4) and had similar characteristics as other affiliates (Section II.D). These results suggest that they were facing the same types of shocks and would have continued to grow in parallel without the lending cut. Third, the findings are robust to several other determinants of affiliate growth, including country-industry-year fixed effects, affiliate size, and financial ratios.

IV. The Effects on Affiliate Growth

We investigate whether the disruption to parents' credit supply, due to Commerzbank's lending cut, affected the growth of international affiliates.

A. Affiliate Sales over Time

We begin by studying affiliate sales over time. Capital constraints can affect sales in the short run through several channels. First, a shock to the availability of working capital may force affiliates to reduce variable inputs and labor, lowering their sales (Chari, Christiano, and Eichenbaum (1995)). Second, a

negative shock to working capital may force firms to halt the transport of products to customers, which immediately harms their sales. Third, firms maintain working capital stocks as liquidity buffers to guard against unexpected shocks (Calomiris, Himmelberg, and Wachtel (1995)). When working capital falls, firms may choose to rebuild their liquidity buffers by halting expenditures that generate sales (e.g., on variable inputs, marketing, and distribution). Finally, working capital may pay for investment in factors of production, like machinery (Fazzari and Petersen (1993)). Sales is also an interesting outcome with respect to aggregate growth because aggregate value added equals sales of final-use products.

We use the following specification for sales y of affiliate a located in country c in year t :

$$\begin{aligned} \log(y_{act}) = & \sum_{\tau=2002}^{2015} \beta_{\tau} \times \text{Parent CB dep} \times \mathbb{1}(t = \tau) + \gamma_a + \lambda_t \\ & + \sum_{\tau=2002}^{2015} \phi'_{\tau} \times \mathbf{X}_{ac} \times \mathbb{1}(t = \tau) + \varepsilon_{act}. \end{aligned} \quad (2)$$

The treatment variable of interest is *Parent CB dep*, measured in 2006, the final year before the U.S. housing market crisis. The coefficients of interest, β_{τ} , measure the effect (in log points) of parent Commerzbank dependence in year τ relative to 2006.

The specification includes affiliate fixed effects γ_a to control for time-invariant differences across affiliates and year fixed effects λ_t to control for macroeconomic shocks. The vector $\mathbf{X}_{ac}$ contains a baseline set of affiliate-level controls, measured in 2006 and interacted with year fixed effects. In the baseline specification, we include fixed effects for size (deciles of sales), industry, country, and deciles of leverage. We include these controls because the existing literature has shown that during our sample period, there were systematic differences in firm growth correlated with firm size (Fort et al. (2013)), industry (Alviarez, Cravino, and Levchenko (2017)), country (Eaton et al. (2016)), and leverage (Giroud and Mueller (2017)). Results are robust to a range of additional specifications that employ alternative definitions of the outcome variable, a balanced panel, country-industry-year fixed effects, and parent characteristics (see Internet Appendix IV). Standard errors are two-way clustered at the level of the country and the parent.

We plot the relationship between affiliate sales and parent Commerzbank dependence for a parent with average Commerzbank dependence, relative to the precrisis baseline year 2006 and conditional on all controls interacted with year fixed effects (Figure 4). The point estimates for the years before 2008 are all close to zero. This result shows that affiliates whose parents had higher Commerzbank dependence were on parallel trends to other affiliates, consistent with the identification assumption.

After Commerzbank cut lending to German parents in 2008, sales of affiliates with greater parent Commerzbank dependence fell sharply, relative to other affiliates. Their sales recovered partially in the subsequent year, but still remained below the levels of other affiliates in 2009 and 2010. From 2011 onward, the levels were similar again. Taken together, the dynamic pattern in the graph suggests that sales of affected affiliates were lower from 2008 to 2010, but recovered in 2011.

B. Affiliate Sales Fell Temporarily

We formally test whether Commerzbank's lending cut lowered the sales of international affiliates temporarily by estimating the effect separately for two periods, 2008 to 2010 and 2011 to 2015:

$$\log(y_{act}) = \beta_1 \times \text{Parent CB dep} \times \mathbb{1}(2008-10)_t + \beta_2 \times \text{Parent CB dep} \times \mathbb{1}(2011-15)_t + \gamma_a + \lambda_t + \phi'_1 \times X_{ac} \times \mathbb{1}(2008-10)_t + \phi'_2 \times X_{ac} \times \mathbb{1}(2011-15)_t + \epsilon_{act}. \quad (3)$$

The specification contains parent Commerzbank dependence and all of the controls interacted with time fixed effects for the two periods.

The point estimate implies that sales of affiliates whose parents had average Commerzbank dependence were on average 9.7 log points lower between 2008 and 2010 (significant at the 5% level), relative to affiliates whose parents had zero Commerzbank dependence (Table II, column (3)). Comparing the relative magnitude of the affiliate effect to the parent effect (from Table IA.IX, column (4)), we find that a shock lowering parent sales by 10% reduced affiliate sales by 6.6%.

The estimate for 2011 to 2015 is small, positive, and statistically insignificant. This finding confirms that affected affiliates recovered to the level of unaffected affiliates after 2011. The point estimates in the specifications with fewer controls are similar (columns (1) and (2)). This result suggests that affiliates with greater parent Commerzbank dependence were not hit by different shocks, in line with the parallel-trends identification assumption. The smaller standard errors in column (3) suggest that the controls soak up a significant amount of noise.

Affiliates and parents with greater parent Commerzbank dependence were not more likely to exit or enter (Table IA.X), consistent with the finding that they recovered in 2011 that persisted thereafter. Similar to the evolution of sales, the employment point estimates suggest a temporary drop of about 4.5 log points from 2008 to 2010 and a recovery after 2011 (Table IA.XI).

C. The Role of Parents' Other Relationship Banks

Some parents that had Commerzbank as a relationship bank also relied on other German relationship banks in 2006. The vast majority of these other banks continued to lend during the crisis (Figure 2). We plot the effects on affiliate sales from 2008 to 2010 separately for five quintiles of parent

Table II
Impact of Parent Commerzbank Dependence on Affiliate Sales

The table reports estimates from OLS panel regressions. The outcome variable is $\log(\text{Affiliate sales})$. *Parent CB dep* is the fraction of the parent's relationship banks that were Commerzbank branches in 2006. Time fixed effects comprise two indicator variables for the periods 2008 to 2010 and 2011 to 2015. The following time-invariant control variables are calculated for affiliates in 2006 and interacted with time fixed effects: industry fixed effects, fixed effects for deciles of total sales, fixed effects for deciles of leverage, and fixed effects for the country of the affiliate. The coefficients are scaled to reflect the effect on an affiliate whose parent had average Commerzbank dependence, which was 0.23. R^2 is within-firm R^2 . Standard errors are two-way clustered at the level of the country and the parent. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations.

Outcome	Affiliate Sales		
	(1)	(2)	(3)
<i>Parent CB dep</i> $\times$ 2008-10	-0.1286 (0.0979)	-0.1403* (0.0806)	-0.0967** (0.0489)
<i>Parent CB dep</i> $\times$ 2011-15	0.0574 (0.0527)	0.0486 (0.0523)	0.0298 (0.0422)
R^2	0.011	0.038	0.092
Number of firms	2,695	2,695	2,695
Observations	24,941	24,941	24,941
Affiliate FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Size bin FE $\times$ Time FE	No	Yes	Yes
Industry FE $\times$ Time FE	No	Yes	Yes
Country FE $\times$ Time FE	No	No	Yes
Leverage bin FE $\times$ Time FE	No	No	Yes

Commerzbank dependence (Figure 5). The excluded bin is for affiliates with zero parent Commerzbank dependence (see Internet Appendix IV for details).

The point estimates for affiliates with parent Commerzbank dependence in the bins up to 0.25 are small and insignificant. In contrast, point estimates are negative, significant, and of roughly equal magnitude for the two bins between 0.25 and 0.5. Finally, the coefficient for the top quintile of affiliates above 0.5 is the largest and also significant. These results show that multinationals were able to substitute missing credit from Commerzbank if they had other, preexisting relationship banks. As a result, the effects on affiliates are not driven by all affiliates whose parents had some dependence on Commerzbank, but rather by affiliates of parents with relatively high dependence.

D. Affiliates Short-Run Inputs and Production Capital

Short-term assets include inputs into short-run production and sales, such as holdings of raw materials and other variable inputs; working capital to pay for employees, suppliers, and the distribution of products to points of sale; trade credit issued to external customers; and liquidity buffers against shocks. We find that affiliates reduced their stock of short-term assets from 2008 to

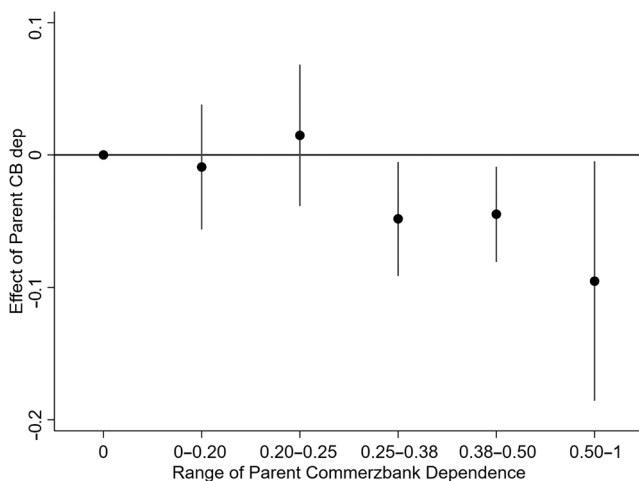


Figure 5. Impact on affiliate sales by bins of parent Commerzbank dependence. The figure plots the relationship between bins of *Parent Commerzbank Dependence* and *Affiliate sales* from 2008 to 2010. The bins represent the quintiles of parent Commerzbank dependence for positive values of parent Commerzbank dependence. The gray-dashed lines represent 90% confidence intervals. The coefficients are estimated in a single regression. The following time-invariant control variables are calculated for affiliates in 2006 and interacted with indicator variables for 2008 to 2010 and 2011 to 2015: industry fixed effects, fixed effects for deciles of total sales, fixed effects for deciles of leverage, and country fixed effects. The specification also contains affiliate and year fixed effects and parent Commerzbank dependence interacted with the 2011 to 2015 indicator. The coefficients are scaled by the average parent Commerzbank dependence, which was 0.23, to make them comparable to the previous results. Standard errors are two-way clustered at the level of the country and the parent. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations.

2010 (Table III, column (1)). The point estimate implies a decrease of 4.9 log points from 2008 to 2010 (significant at the 5% level). Short-term assets fully recovered after 2011 (insignificant point estimate of 0.002).

The reduction in short-term assets is consistent with the view that affiliates became capital-constrained after Commerzbank's lending cut, which led them to reduce short-run inputs. Accordingly, the evolution of short-term assets over time mirrored the evolution of sales, with a reduction from 2008 to 2010 and full recovery thereafter.

We show that production capital (including tangible and intangible assets) was 2 log points lower from 2008 to 2010, but the effect is imprecisely estimated (column (2)). The slightly weaker point estimate for production capital, relative to short-term assets, may be due to the fact that affiliates do not typically engage in large investment projects and innovation on their own, so that affiliates did not have many potential projects that they could reduce in the first place (Gumpert et al. (2023)). In addition, in response to a temporary shock, firms may find it optimal to reduce short-term assets, which primarily harms short-run sales, by more than production capital, which can harm sales over a longer period of time (see Mezzanotti and Simcoe (2022)).

Table III
Impact of Parent Commerzbank Dependence on Affiliate Assets, Liabilities, and Equity

The table reports estimates from OLS panel regressions. The outcome is $\log(\text{Short-term assets of the affiliate excluding short-term claims on the parent by the affiliate})$ in column (1), $\log(\text{Production capital})$ in column (2), $\log(\text{Long-term loans of the affiliate excluding long-term loans to the parent})$ in column (3), $\log(\text{Liabilities of the affiliate excluding liabilities owed to the parent})$ in column (4), and $\log(\text{Equity invested by nonparents})$ in column (5). The regressors are explained in Table II, and the controls include fixed effects for size bin, industry, country, and leverage bin. The coefficients are scaled to reflect the effect on an affiliate whose parent had average Commerzbank dependence, which was 0.23. R^2 is within-firm R^2 . Standard errors are two-way clustered at the level of the country and the parent. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations.

Outcome	Short-Term Assets (1)	Production Capital (2)	External LT Loans (3)	External Liabilities (4)	External Equity (5)
<i>Parent CB dep × 2008-10</i>	−0.0487** (0.0197)	−0.0196 (0.0322)	0.0898 (0.0774)	0.0008 (0.0349)	0.0153 (0.0441)
<i>Parent CB dep × 2011-15</i>	0.0024 (0.0337)	−0.0228 (0.0415)	0.0504 (0.0829)	0.0289 (0.0499)	0.0024 (0.0526)
R^2	0.089	0.064	0.047	0.067	0.074
Number of firms	2,695	2,695	2,695	2,695	2,695
Observations	24,941	24,941	24,941	24,941	24,941
Affiliate FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Controls × Time FE	Yes	Yes	Yes	Yes	Yes

E. Affiliates Did Not Raise External Financing

We find that affiliates did not reduce external long-term loans to nonparents (Table III, column (3)). This would have required early repayment of long-term loans by external borrowers, which is difficult to enforce.

There was also no change in affiliates' external liabilities, which includes borrowing from banks, and in equity raised from nonparents (columns (4) and (5)). Both coefficients are close to zero and insignificant. These estimates suggest that, on average, affiliates were not able to use external funding to overcome their capital constraints. This view is supported by a large literature showing that firms in many countries are borrowing-constrained and unable to raise additional external financing in the short term when hit by a shock (for overviews, see Banerjee and Duflo (2014), Catherine et al. (2022)), likely due to asymmetric information problems.

V. The Importance of Internal Capital Markets

This section shows that internal capital markets responded to the shock to parents' credit supply and that real effects were concentrated among affiliates that increased long-term lending to the parent.

A. Affiliates with Greater Long-Term Loans to the Parent

We conduct a heterogeneity analysis. We use preexisting internal capital positions to assign affiliates to two heterogeneity categories. Using preexisting positions is preferable to using actual changes in internal capital flows because actual changes are endogenous to other shocks hitting affiliates.

The first heterogeneity characteristic we analyze is whether affiliates had issued long-term loans (maturity above one year) to their parent at some point before 2008. Parents had already set up a financial infrastructure to borrow from these affiliates. This made it easier for parents to quickly borrow capital from these affiliates after Commerzbank's lending cut. We test for heterogeneous treatment effects by specifying:

$$\begin{aligned} \log(y_{act}) = & (\beta_1 + \beta'_1 \times het_{ac}) \times Parent\ CB\ dep \times \mathbb{1}(2008-10)_t \\ & + \beta_2 \times Parent\ CB\ dep \times \mathbb{1}(2011-15)_t + \gamma_a + \lambda_t \\ & + \phi'_1 \times X_{ac} \times \mathbb{1}(2008-10)_t + \phi'_2 \times X_{ac} \times \mathbb{1}(2011-15)_t + \epsilon_{act}. \end{aligned} \quad (4)$$

The indicator variable het_{ac} identifies whether an affiliate falls into the given heterogeneity category. The coefficient β_1 estimates the effect of parent Commerzbank dependence from 2008 to 2010 on affiliates that do not fall into the given heterogeneity category, while β'_1 measures the additional effect from 2008 to 2010 on affiliates that fall into the category. We add het_{ac} to the control variables, interacted with the indicator for 2008 to 2010. This avoids bias if there were shocks that affected all affiliates in the given heterogeneity category independent of their parent Commerzbank dependence.

In a specification without the heterogeneity term, we find a positive effect on internal loans by affiliates to the parent. Affiliates with average parent Commerzbank dependence increased long-term loans by 8.7 log points from 2008 to 2010 (significant at the 10% level), relative to affiliates with zero parent Commerzbank dependence (Table IV, column (1)). Since the average parent owned 88% of an affiliate, parents were likely able to direct affiliates to increase internal lending. In column (2), we include the heterogeneity term. We find that the effect on long-term loans was driven by affiliates with previous long-term loans before 2008. The coefficient on the interaction between previous loans and parent Commerzbank dependence is positive, large, and significant at the 5% level. Affiliates that had previously lent to their parent increased internal lending to their parent by more from 2008 to 2010.

Next, we examine whether the sales decrease was stronger for affiliates that increased lending by more. We find that the additional sales reduction for affiliates with previous long-term loans was large and significant at the 1% level (Table V, column (1)). The effect on affiliates without previous long-term loans was smaller and statistically insignificant. This finding implies that the effect on affiliate sales was to a large extent driven by affiliates that increased internal lending to their parent after the lending cut.

We consider whether other characteristics of affiliates with previous long-term loans can explain this finding. Affiliates with previous loans had similar

Table IV
Impact of Parent Commerzbank Dependence on Internal Capital
Market Flows

The table reports estimates from OLS panel regressions. The outcome in columns (1) and (2) is $\log(\text{Long-term loans by the affiliate to the parent})$. The outcome in columns (3) and (4) is $\log(\text{Short-term claims on the parent by the affiliate})$. Column (2) analyzes heterogeneity by whether the affiliate had issued long-term loans to its parent in any year before 2008. Column (4) analyzes heterogeneity by whether the affiliate held short-term claims on its parent in any year before 2008. The remaining regressors are explained in Table II, and the baseline controls include fixed effects for size bin, industry, country, and leverage bin. The coefficients are scaled to reflect the effect on an affiliate whose parent had average Commerzbank dependence, which was 0.23. R^2 is within-firm R^2 . Standard errors are two-way clustered at the level of the country and the parent. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations.

Outcome	LT Loans to Parent		ST Claims on Parent	
	(1)	(2)	(3)	(4)
Parent CB dep $\times$ 2008-10	0.0867* (0.0508)	0.0544 (0.0340)	-0.1878** (0.0929)	-0.0843 (0.0690)
Parent CB dep $\times$ 2008-10 $\times$ Previous long-term loans to parent		0.5427** (0.2206)		
Parent CB dep $\times$ 2008-10 $\times$ Previous short-term claims on parent				-0.1938* (0.1094)
Parent CB dep $\times$ 2011-15	0.1015 (0.0696)	0.0985 (0.0689)	-0.1437 (0.1062)	-0.1418 (0.1058)
R^2	0.031	0.035	0.059	0.059
Number of firms	2,695	2,695	2,695	2,695
Observations	24,941	24,941	24,941	24,941
Affiliate FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls $\times$ Time FE	Yes	Yes	Yes	Yes
Previous long-term loans to parent $\times$ Time FE	No	Yes	No	No
Previous short-term claims on parent $\times$ Time FE	No	No	No	Yes

size and pregrowth to other affiliates (Table IA.XII). Parents held a marginally larger ownership share in these affiliates and these affiliates were slightly more levered, although the point estimates are small. This is consistent with the view that parents are able to withdraw internal lending from affiliates over which they have more control and that these affiliates become marginally more levered to lend to the parent. However, the effects on affiliate growth are stable when conditioning on all of these characteristics. We also find no evidence that the heterogeneous effects with respect to previous long-term loans are driven by leverage or short-term claims on the parent (relative to affiliate assets).

B. Affiliates with Greater Short-Term Claims on the Parent

The second heterogeneity characteristic we consider relates to internal trade. We measure whether affiliates held internal short-term claims

Table V
Effects were Stronger for Affiliates with Large Internal Capital Market Positions

The table reports estimates from OLS panel regressions. The outcome variable is $\log(\text{Affiliate sales})$. Column (1) analyzes heterogeneity by whether the affiliate had issued long-term loans to its parent in any year before 2008. Column (2) analyzes heterogeneity by whether the affiliate held short-term claims on its parent in any year before 2008. The remaining regressors are explained in Table II, and the baseline controls include fixed effects for size bin, industry, country, and leverage bin. The coefficients are scaled to reflect the effect on an affiliate whose parent had average Commerzbank dependence, which was 0.23. R^2 is within-firm R^2 . Standard errors are two-way clustered at the level of the country and the parent. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations.

Outcome	Affiliate Sales		
	(1)	(2)	(3)
Parent CB dep $\times$ 2008-10	-0.0647 (0.0408)	-0.0305 (0.0434)	-0.0265 (0.0415)
Parent CB dep $\times$ 2008-10 $\times$ Previous long-term loans to parent	-0.5414*** (0.1774)		-0.5094*** (0.1824)
Parent CB dep $\times$ 2008-10 $\times$ Previous short-term claims on parent		-0.1244* (0.0668)	-0.0751 (0.0604)
Parent CB dep $\times$ 2011-15	0.0327 (0.0431)	0.0310 (0.0423)	0.0333 (0.0430)
R^2	0.093	0.092	0.093
Number of firms	2,695	2,695	2,695
Observations	24,941	24,941	24,941
Affiliate FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls $\times$ Time FE	Yes	Yes	Yes
Previous long-term loans to parent $\times$ 2008-10	Yes	No	Yes
Previous short-term claims on parent $\times$ 2008-10	No	Yes	Yes

(maturity less than a year) on the parent before 2008. Internal short-term claims are largely composed of internal trade credit to the parent. Hence, internal short-term claims are a commonly used proxy for internal trade from affiliates to the parent.¹⁶

Without testing for heterogeneity, we find that affiliates with average parent Commerzbank dependence reduced short-term claims by 18.8 log points from 2008 to 2010 (significant at the 5% level), relative to affiliates with zero parent Commerzbank dependence (Table IV, column (3)). A likely explanation is that financially constrained parents demanded fewer production inputs from affiliates. This slowed internal trade and therefore short-term claims by affiliates

¹⁶ See, for example, Overesch (2006). We do not observe internal trade credit directly in the MiDi data on affiliates. But we can observe both internal trade credit and internal short-term claims in the Ustan data on German multinationals. The correlation between the percent change in trade credit and the percent change in short-term claims (both to affiliates in the same multinational) is 0.93 for the year before the lending cut. This confirms that changes in internal short-term claims are a good predictor of changes in internal trade credit.

on their parent. The slowdown in short-term claims was larger for affiliates with previous short-term claims. The interaction between previous short-term claims and parent Commerzbank dependence is negative and significant at the 10% level (column (4)). This confirms that affiliates with previous short-term claims were more exposed to a slowdown in internal trade with the parent. We also find that the effect on sales was larger for affiliates with previous short-term claims (significant at the 10% level), relative to affiliates without short-term claims (Table V, column (2)). This suggests that the reduction in internal trade played a role in lowering affiliate sales.

C. The Importance of Long-Term Loans to the Parent

To compare the effects of internal lending and internal trade, we include both heterogeneity categories in one specification. The additional effect on affiliates with previous long-term loans remains large and significant at the 1% level (Table V, column (3)). The additional effect on affiliates with previous short-term claims becomes insignificant, but remains of similar magnitude. These estimates are consistent with the view that greater internal lending contributed to the transmission of the lending cut to affiliates, while internal trade may have also played a role.

For completeness, we also examine the remaining observable internal capital positions. We find that equity invested by the parent into the affiliate and liabilities owed to the parent did not change significantly from 2008 to 2010 (Table IA.XIII).¹⁷ A likely reason is that equity divestment is a relatively slow and cumbersome process. Moreover, affiliates in which the parent held a larger equity share did not experience faster drops in equity and sales (Table IA.XIV). Taken together, the evidence suggests that affiliates primarily transferred capital to their parent by increasing internal lending.

VI. Mechanisms and Frictions Driving Internal Capital Market Flows

In this section, we analyze which mechanisms and frictions determine flows through internal capital markets and their real effects.

A. “Darwinism” among International Affiliates

In standard models in which parent managers maximize total profits of the multinational, parent managers are “Darwinist” and allocate a higher level of capital to affiliates with high growth potential (Stein (1997), Inderst and Mueller (2003)). In such models, a negative credit shock to the parent has weaker effects on affiliates with high growth potential, as we show theoretically in Internet Appendix II. However, in models in which affiliates engage in rent-seeking (Scharfstein and Stein (2000)) or power struggles (Rajan,

¹⁷ Deutsche Bundesbank data do not differentiate between long- and short-term liabilities owed to parents.

Table VI
Mechanisms and Frictions Driving Internal Capital Market Flows

The table reports estimates from OLS panel regressions. The outcome variable is $\log(\text{Affiliate sales})$. Column (1) tests for heterogeneity in affiliate sales growth from 2006 to 2007. Column (2) tests for heterogeneity in whether the affiliate host country was in the top or bottom 10% of the distribution of credit to GDP in 2006 (based on World Bank data). Columns (3) to (5) analyze heterogeneity in whether the affiliate was located in Asia, the EU, or the United States. The remaining regressors are explained in Table II, and the baseline controls include fixed effects for size bin, industry, country, and leverage bin. The coefficients are scaled to reflect the effect on an affiliate whose parent had average Commerzbank dependence, which was 0.23. R^2 is within-firm R^2 . Standard errors are two-way clustered at the level of the country and the parent. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, own calculations.

Outcome	Affiliate Sales				
	(1)	(2)	(3)	(4)	(5)
Parent CB dep $\times$ 2008-10	-0.0953* (0.0490)	-0.1080* (0.0596)	-0.1054** (0.0420)	-0.0748 (0.0582)	-0.0913* (0.0539)
Parent CB dep $\times$ 2008-10 $\times$ Affiliate pregrowth	0.0597** (0.0280)				
Parent CB dep $\times$ 2008-10 $\times$ Credit/GDP low		-0.1984*** (0.0620)			
Parent CB dep $\times$ 2008-10 $\times$ Credit/GDP high		0.0276 (0.0550)			
Parent CB dep $\times$ 2008-10 $\times$ Affiliate in Asia			0.0525 (0.1122)		
Parent CB dep $\times$ 2008-10 $\times$ Affiliate in EU				-0.0393 (0.0618)	
Parent CB dep $\times$ 2008-10 $\times$ Affiliate in US					-0.0550 (0.0519)
Parent CB dep $\times$ 2011-15	0.0333 (0.0427)	0.0298 (0.0430)	0.0302 (0.0425)	0.0301 (0.0423)	0.0300 (0.0422)
R^2	0.117	0.073	0.092	0.092	0.092
Number of firms	2,244	2,661	2,695	2,695	2,695
Observations	23,143	24,681	24,941	24,941	24,941
Affiliate FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Controls $\times$ Time FE	Yes	Yes	Yes	Yes	Yes
Affiliate pregrowth $\times$ 2008-10	Yes	No	No	No	No

Servaes, and Zingales (2000)), parent managers may be relatively “Socialist” and treat all affiliates equally (Matvos and Seru (2014)).

We examine which model applies to multinationals by testing heterogeneity with respect to affiliates’ precrisis sales growth from 2006 to 2007. We find that affiliates with higher precrisis growth were less affected by their parent’s Commerzbank dependence (Table VI, column (1)). The interaction effect between precrisis growth and parent Commerzbank dependence is positive and significant at the 5% level. This suggests that parent managers treat international affiliates in a “Darwinist” manner, so that international affiliates with high growth potential are harmed less.

B. Home Bias and “Socialism” among Affiliates within Germany

We examine whether German multinationals also transmitted negative effects to their domestic affiliates located within Germany. The Bundesbank data only cover international affiliates, so we construct a new data set of affiliates that were located in Germany and owned by a German multinational parent. Specifically, we use Orbis Historical Financials to identify which German firms were at least 50% owned by another German firm that also had a foreign affiliate in 2007, the first year with ownership data in Orbis. We use the data to test whether (i) German affiliates were harmed less on average and (ii) whether weak German affiliates were affected more strongly, as was the case for weak international affiliates.

First, we find a negative coefficient on parent Commerzbank dependence in a sample of only German affiliates (Table VII, column (1)). However, the estimate is close to zero and statistically insignificant. This finding suggests that managers of German multinationals have home bias, so they disproportionately transmit harmful shocks to international affiliates rather than domestic affiliates.

Second, the interaction effect between precrisis growth and parent Commerzbank dependence is statistically insignificant (Table VII, column (2)). If anything, the sign is negative, implying that weak affiliates were not harmed more strongly. This finding implies that there is “Socialism” within Germany, whereby affiliates located in Germany are treated equally independent of their growth potential.¹⁸

Using a structural model, Matvos and Seru (2014) find that managers of U.S. conglomerates have a preference for “Socialism,” which means that, on average, they allocate more capital to weaker divisions. Our findings enrich this picture. We find that the dynamic response to crises is also “Socialist” in the parent’s home country, as parent managers treat domestic affiliates equally in response to a negative shock. In contrast, parent managers are “Darwinist” with respect to international affiliates. A likely explanation for this divergence between domestic and international affiliates is that parent managers are more exposed to power struggles and rent-seeking by domestic affiliates, whereas they can engage in relatively unobstructed winner picking among international affiliates.

C. Access to Local Credit Markets Dampened Negative Effects

We next analyze whether frictions in local credit markets shaped the effects on affiliates, as predicted by the theoretical model in Internet Appendix II. We find that the sales effect was indeed stronger for affiliates in countries

¹⁸ In Table IA.XV, we show that the results for domestic affiliates are statistically different relative to those for international affiliates, combining the Orbis data on domestic affiliates with the Bundesbank MiDi data on international affiliates. In addition, we find that the direct Commerzbank dependence of domestic affiliates reduced affiliate sales (Table VII, column (3)), in line with Huber (2018).

Table VII
Impact on Sales of Affiliates within Germany

The table reports estimates from OLS panel regressions. The sample only contains affiliates that were located within Germany, were owned at least 50% by a German parent, and whose parent had at least one foreign affiliate. The outcome is $\log(\text{Affiliate sales})$. The regressors are explained in Table II, and the controls include fixed effects for size bin, industry, location, and leverage bin. *Other affiliates' CB dep* is the average Commerzbank dependence of other affiliates within Germany owned by the same parent. The coefficients are scaled to reflect the effect on a firm with average Commerzbank dependence, which was 0.23. R^2 is within-firm R^2 . Standard errors are clustered at the level of the parent. Data Source: Orbis Historical Ownership 2007, Orbis Historical Financials 1990 to 2015, own calculations.

Outcome	Affiliate Sales		
	(1)	(2)	(3)
<i>Parent CB dep</i> $\times$ 2008-10	-0.0118 (0.0355)	-0.0048 (0.0368)	-0.1212 (0.0840)
<i>Parent CB dep</i> $\times$ 2008-10 $\times$ <i>Affiliate pregrowth</i>		-0.0825 (0.0554)	-0.0417 (0.0735)
<i>Affiliate CB dep</i> $\times$ 2008-10			-0.0970 ^{***} (0.0402)
<i>Affiliate CB dep</i> $\times$ 2008-10 $\times$ <i>Affiliate pregrowth</i>			0.0888 (0.0789)
<i>Other affiliates' CB dep</i> $\times$ 2008-10			0.1235 (0.0781)
<i>Other affiliates' CB dep</i> $\times$ 2008-10 $\times$ <i>Affiliate pregrowth</i>			-0.0123 (0.0618)
<i>Parent CB dep</i> $\times$ 2011-15	-0.0143 (0.0431)	-0.0166 (0.0434)	-0.1991 [*] (0.1033)
<i>Affiliate CB dep</i> $\times$ 2011-15			-0.0269 (0.0375)
<i>Other affiliates' CB dep</i> $\times$ 2011-15			0.1312 (0.0804)
R^2	0.230	0.233	0.263
Number of firms	940	940	550
Observations	11,430	11,430	6,473
Affiliate FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls $\times$ Time FE	Yes	Yes	Yes
Affiliate pre-growth $\times$ 2008-10	No	Yes	Yes

with underdeveloped credit markets, as measured by the ratio of credit to GDP (Table VI, column (2)). When affiliates found it harder to borrow locally, they were less able to fund internal lending with external funds and instead had to reduce their sales more.

However, even in developed credit markets, affiliates still suffered significant sales losses, as indicated by the stable and significant baseline coefficient (column (2)). This is consistent with existing work showing that firms are borrowing-constrained even in economies with large credit markets, at least in the short run (Lian and Ma (2020)). Overall, frictions in the access to

capital determine the extent to which internal capital market shocks lead to real effects.¹⁹

D. Currency, Geography, Local Business Cycles, and Capital Controls

We next study whether frictions due to broad regional differences affected shock transmission. On the one hand, we may expect stronger effects for affiliates in EU countries, since it is easier to transfer capital across countries sharing the same currency and regulatory environment. On the other hand, parent managers may display proximity bias for geographically and culturally close countries, which would reduce the effect on EU affiliates. On net, we find no significant difference for affiliates located in Asia, the EU, and the United States (Table VI, columns (3) to (5)). In Internet Appendix V, we also report no significant differences by GDP per capita growth, house price growth, or corporate tax rates in the affiliate host country.

We do not find that capital controls in affiliate host countries had strong effects (Table IA.XVII), using three different capital control measures developed by Fernández et al. (2016). A likely reason is that capital flows within firms are difficult to monitor and prevent, so most countries do not explicitly target internal capital markets with capital control measures. Overall, parents were seemingly able to allocate capital relatively freely across the world.

E. The Role of Internal Trade and Reselling

A large literature in international economics studies internal trade between affiliates and parents (e.g., Alfaro, Kalemli-Özcan, and Volosovych (2007), Alfaro and Charlton (2009), Ramondo, Rappoport, and Ruhl (2016)). We present several pieces of evidence that internal trade explains part, but not the entire effect on affiliate sales, suggesting that internal capital markets also played an important role.

If only internal trade mattered, we would expect weaker effects for horizontal and service sector affiliates (Cravino and Levchenko (2017)). We first analyze horizontal affiliates, which operated in the same industry as their parent and were therefore less likely to trade internally with the parent. We find no significant difference between the effect on horizontal and vertical affiliates (Table IA.XVIII, column (1)).²⁰ We also find no significant difference for affiliates in the service sector, which are less likely to trade with their parent (column (2)). Finally, the coefficient remains similar when we drop affiliates in wholesale and retail (Table IA.XIX), suggesting that the effect is not driven just by affiliates that resell their parent's output.

¹⁹ We also find that larger affiliates suffered smaller effects (Table IA.XVI), consistent with the view that larger firms can access external finance more easily (Farre-Mensa and Ljungqvist (2016)), but the result is not statistically significant.

²⁰ We use one-digit industries to define horizontal affiliates, but find similar results using two- or three-digit industries.

F. Affiliates Did Not Directly Borrow from Commerzbank

Affiliates' direct exposure to Commerzbank branches (in Germany or abroad) cannot explain the effects on affiliate growth. First, affiliate liabilities toward banks and other nonparents did not change significantly, suggesting that affiliates were not directly exposed to a bank lending cut (Table III, column (4)). Moreover, we find that the correlation between parent and affiliate Commerzbank dependence is 0.025 in Orbis and that only 3% of foreign affiliates had a direct bank relationship to Commerzbank. (We cannot match affiliate relationship banks to MiDi directly.) Finally, we find similar results when excluding all countries in which Commerzbank had a foreign branch and when controlling for the fraction of bank lending in the affiliate host country that came directly from German branches of the three largest German banks in 2006 (Table IA.XX).

VII. Comparison: The Transmission of a Nonfinancial Flooding Shock

We analyze how internal capital markets and affiliates respond to a different, nonfinancial shock that hit parents. This analysis helps us to understand whether internal capital markets operate differently in the case of nonfinancial shocks, relative to financial shocks like Commerzbank's lending cut.

A. The Flood Lowered Parent Assets But Not Credit Supply

We identify parents that were exposed to the large-scale German floods of June 2013 but whose relationship banks were still able to lend to them. The floods affected regions around the rivers Elbe and Danube and were caused by unexpectedly heavy rainfall. The floods covered the largest geographic area of any flood in Germany since the start of official records in 1950. As argued by Koetter, Noth, and Rehbein (2020), banks located outside flooded regions provided "recovery lending" to flooded firms, whereas some banks located in flooded regions were directly harmed by flooding. To isolate a nonfinancial shock, we therefore define a treatment indicator for "flooded parents" with stable credit supply. We include parents with (i) a headquarter in flooded states (where at least 0.24% of flood insurance contracts were activated according to the German Association of Insurers) and (ii) a relationship bank outside these states (according to the Creditreform data on relationship banks).

We first examine the direct impact of the flood on parents, using an analogous specification as for the analysis of Commerzbank's lending cut. We find that flooded parents experienced an average reduction in assets of approximately 2 log points (Table IA.XXI, column (1)) in the period 2013 to 2015, relative to nonflooded parents. The reduction is consistent with the flood destroying production capital and inventory, which takes time to rebuild. However, treated parents were not financially constrained, as their bank debt increased by 20.6 log points (column (2)), a magnitude roughly equal to the decline in bank debt following Commerzbank's lending cut. This finding suggests that

Table VIII
Impact of the 2013 Flood on Affiliates

The table reports estimates from OLS panel regressions. The outcome in column (1) is $\log(\text{Long-term loans by the affiliate to the parent})$. The outcome in column (2) is $\log(\text{Short-term claims on the parent by the affiliate})$. The outcome variable in columns (3) and (4) is $\log(\text{Affiliate sales})$. Column (1) analyzes heterogeneity by whether the affiliate had issued long-term loans to its parent in any year before 2013. Column (2) analyzes heterogeneity by whether the affiliate held short-term claims on its parent in any year before 2013. The remaining regressors are explained in Table II, and the baseline controls include fixed effects for size bin, industry, country, and leverage bin. The coefficients are scaled to reflect the effect on an affiliate whose parent had average Commerzbank dependence, which was 0.23. R^2 is within-firm R^2 . Standard errors are two-way clustered at the level of the country and the parent. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, German Association of Insurers, own calculations.

Outcome	LT Loans to Parent (1)	ST Claims on Parent (2)	Affiliate Sales (3)	Affiliate Sales (4)
<i>Flooded parent 2013 × 2013-2015</i>	−0.0079 (0.0142)	0.0856** (0.0372)	0.0033 (0.0264)	−0.0104 (0.0390)
<i>Flooded parent 2013 × 2013-2015 × Previous long-term loans to parent</i>	−0.0570 (0.1293)			0.0370 (0.1598)
<i>Flooded parent 2013 × 2013-2015 × Previous short-term claims on parent</i>		−0.0330 (0.0564)		0.0063 (0.0432)
R^2	0.018	0.043	0.046	0.047
Number of firms	1,948	1,948	1,948	1,948
Observations	19,012	19,012	19,012	19,012
Affiliate FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls × Time FE	Yes	Yes	Yes	Yes
Previous long-term loans on parent × Time FE	Yes	No	No	Yes
Previous short-term claims on parent × Time FE	No	Yes	No	Yes

treated parents experienced a nonfinancial, asset-destroying shock, but were able to receive strong financing from their German relationship banks.

B. The Effect of the Flood on International Affiliates and Internal Capital Markets

We turn to results at the affiliate level. Affiliates with flooded parents did not increase long-term lending to their flooded parents (Table VIII, column (1)), as parents were able to access capital from their German banks instead. Affiliates slightly increased short-term claims on their parent (column (2)), possibly because affiliates delivered more goods to their parent as the parent was rebuilding production capital and inventory. There is no evidence that affiliate sales fell, as the coefficient is small and insignificant (column (3)). There is also no significant heterogeneity for affiliates with greater internal capital market links to their parent (column (4)).

Taken together, the results suggest that internal capital markets did not transmit the nonfinancial flood shock from parents to international affiliates. This conclusion stands in sharp contrast to the international transmission of Commerzbank's lending cut through internal capital markets. The difference is that parents were financially constrained following the Commerzbank shock, leading them to withdraw capital from their affiliates and thereby harming affiliates' real growth. In contrast, parents were able to rely on German banks for funding after the floods without gathering capital from affiliates.

The evidence implies that multinationals transmit crises forcefully across the globe when crises have a financial component, that is, when directly shocked firms cannot smooth the shock themselves by relying on their own external credit supply. Of course, many crises are both financial and nonfinancial in nature, including the crises studied in Boehm, Flaaen, and Pandalai-Nayar (2019) and Bena, Dinc, and Erel (2022). In this section, we instead focused on a nonfinancial flooding shock. This focus may explain why the estimated transmission effects of the flooding shock are small relative to these other crises and relative to Commerzbank's lending cut.

VIII. The Magnitude of Shock Transmission through Multinationals

In the final analysis, we compare the magnitude of the affiliate-level results to aggregate outcomes in different countries. We calculate in which countries the shock to Commerzbank had the largest effects because it was transmitted through multinationals.

A. The Transmission of Commerzbank's Lending Cut across Countries

We estimate the reduction in total sales of affiliates whose parents were hit by Commerzbank's lending cut, separately for different countries. Specifically, we calculate how much greater total annual sales of German affiliates would have been between 2008 and 2010 if all German affiliates had grown at the same rate as unaffected German affiliates (with zero parent Commerzbank dependence). We call this quantity the "impact on total sales of German affiliates." For country c , this measure is given by:

$$\begin{aligned} (\text{Impact on total sales of German affiliates})_c = & \\ |\hat{\beta}_1| \times (\text{Weighted average of parent CB dep of affiliates of German parents})_c & \\ \times (\text{Total sales of affiliates of German parents in 2006})_c. & \end{aligned} \quad (5)$$

Estimate $\hat{\beta}_1$ is the effect of parent Commerzbank dependence on the sales of an individual affiliate from 2008 to 2010. We use the baseline estimate $\hat{\beta}_1$ from Table II, column (3) for this calculation. To capture the parent Commerzbank dependence of a representative German affiliate in country c , we use the sales-weighted average parent Commerzbank dependence of German affiliates. The product of the first two terms is an estimate of the sales loss

Table IX
The Impact in Different Countries

We calculate the “impact” on total annual sales of German affiliates between 2008 and 2010 if all German affiliates had grown at the same rate as unaffected German affiliates (with zero parent Commerzbank dependence). The table reports the impact scaled by aggregate sales of nonfinancial firms in the given country. The mean and median refer to the countries listed in the table. We include the most common locations of German affiliates, apart from China and Switzerland because data on total sales of German affiliates are not readily available. Data Source: Research Data and Service Centre of the Deutsche Bundesbank, MiDi 2002 to 2015, Eurostat, U.S. Census Bureau, own calculations.

Affiliate Country	
Czech Republic	0.49
Austria	0.36
Poland	0.31
Netherlands	0.10
Spain	0.10
France	0.07
The United Kingdom	0.07
Italy	0.05
The United States	0.03
Mean	0.18
Median	0.10

(in percent) at a representative German affiliate in country c . Finally, the third term is the total value of sales of nonfinancial German affiliates in country c in 2006, calculated using MiDi. By multiplying the effect on a representative affiliate with the third term, we get an estimate of the impact on total sales of German affiliates in country c , relative to what total sales would have been had all German affiliates grown at the same rate as unaffected affiliates. We are interested in how large the impact was compared to the aggregate economy, so we report the impact as a percentage of aggregate sales.

We present results for the most common locations of German affiliates in Table IX. The mean impact relative to aggregate sales was 0.18%. It was modest in the United States (0.03%), but large in the Czech Republic (0.49%), Austria (0.36%), and Poland (0.31%). The differences across countries are driven mostly by heterogeneity in the size of German affiliates relative to the aggregate economy. German affiliates were responsible for only 0.2% of aggregate sales in the United States, but for a large part of sales in economies at Germany’s Eastern border, up to 4% in the Czech Republic.

To illustrate the magnitude of the effects, consider the example of the Czech Republic. If affected German affiliates (with positive Commerzbank dependence) had grown at the same rate as unaffected affiliates, total sales of affected German affiliates in the Czech Republic between 2008 and 2010 would have been 1.8 billion euro higher (or 0.49% of aggregate sales). Hence, Commerzbank’s lending cut, even though it initially hit only German firms, was

an important determinant of the distribution of firm growth in the Czech Republic.²¹

Previous research suggests that large firms or banks affect growth in their home country. In our setting, a financial shock originating from one bank in one country (Commerzbank in Germany) had large effects on the growth of firms in other countries.

B. Aggregate Implications

Our discussion of Table IX so far has measured the impact on affected affiliates (with positive parent Commerzbank dependence), relative to the growth of unaffected affiliates. These impact numbers would be identical to aggregate effects if the growth of unaffected affiliates had been the same with and without the sales losses at affected affiliates. General equilibrium effects could cause the growth of unaffected affiliates to differ across the two cases. On the one hand, sales of unaffected firms may have increased when affected affiliates became constrained, as product demand shifted from affected to unaffected affiliates. On the other hand, sales of unaffected firms may have fallen because the constraints at affected affiliates lowered aggregate demand and propagated through input-output links.

The impact numbers may be informative about the order of magnitude of aggregate effects, according to recent work. For instance, Huber (2018) estimates that general equilibrium effects of a credit shock exacerbate losses at the regional level. This suggests that aggregate losses may be even larger than the impact numbers in Table IX. Indeed, calibrated models in Chodorow-Reich (2014), Herreño (2020), and Sraer and Thesmar (2023) suggest that general equilibrium effects are likely to harm the growth of unaffected affiliates or have at most a weakly positive effect.

IX. Conclusion

This paper shows that a shock to one bank can have far-reaching effects that extend beyond countries directly borrowing from the shocked bank. We identify a shock to Commerzbank in Germany. This shock had direct effects on Commerzbank's borrowers in Germany because Commerzbank cut their loan supply. However, the shock ultimately also affected affiliates of German multinationals in other countries, even though the affiliates were not directly borrowing from Commerzbank.

The analysis produces new insights on the workings of internal capital markets. We analyze a detailed regulatory data set that reports internal capital market positions between German multinational parents and each international affiliate. We show that the international affiliates financially supported

²¹ All of these effects relate only to transmission through multinationals. Other transmission channels may have further depressed economies abroad because of Commerzbank's lending cut, for example, if German customers demanded fewer goods from abroad. We intentionally hold these other channels constant in our research design, so that we can isolate the effect of multinationals.

their constrained parents through internal lending. The affiliates became financially constrained themselves, experienced lower sales and employment growth, and took three years to recover. Real effects primarily affected affiliates that increased internal lending after the shock, implying that internal capital markets contributed to shock transmission.

Managers allocated the harmful effects across international affiliates in a relatively strategic, “Darwinist” manner, without being limited by frictions due to currencies, geographic distance, or capital controls. However, managers were “Socialist” toward domestic affiliates within Germany and displayed home bias favoring German affiliates. Moreover, international affiliates with access to developed credit markets were able to dampen the effects of the internal capital shock. While the literature has suggested that internal capital flows help affiliates in crisis, these findings show that internal capital markets can also have severe negative impacts on real outcomes.

Aggregating the real effects, we calculate that the transmission of Commerzbank’s lending cut through multinationals had substantial effects in foreign countries. For example, the total impact was roughly 0.4% of aggregate sales in Austria and the Czech Republic and 0.1% in Spain and the United Kingdom. In contrast, we find that internal capital markets did not significantly transmit a nonfinancial flooding shock to international affiliates. These findings suggest that internal capital markets transmit financial shocks, which harm credit access, particularly strongly.

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REFERENCES

- Acemoglu, Daron, Vasco M. Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi, 2012, The network origins of aggregate fluctuations, *Econometrica* 80, 1977–2016.
- Acharya, Viral V., and Philipp Schnabl, 2010, Do global banks spread global imbalances? Asset-backed commercial paper during the financial crisis of 2007–09, *IMF Economic Review* 58, 37–73.
- Alfaro, Laura, and Andrew Charlton, 2009, Intra-industry foreign direct investment, *American Economic Review* 99, 2096–2119.
- Alfaro, Laura, Manuel García-Santana, and Enrique Moral-Benito, 2021, On the direct and indirect real effects of credit supply shocks, *Journal of Financial Economics* 139, 895–921.
- Alfaro, Laura, Şebnem Kalemli-Özcan, and Vadym Volosovych, 2007, Capital Controls and Capital Flows in Emerging Economies: Policies, Practices and Consequences, pp. 19–72 (University of Chicago Press, Chicago, IL).
- Almeida, Heitor, Chang-Soo Kim, and Hwanki Brian Kim, 2015, Internal capital markets in business groups: Evidence from the Asian financial crisis, *Journal of Finance* 70, 2539–2586.
- Alviarez, Vanessa, Javier Cravino, and Andrei A. Levchenko, 2017, The growth of multinational firms in the great recession, *Journal of Monetary Economics* 85, 50–64.
- Amiti, Mary, and David E. Weinstein, 2018, How much do idiosyncratic bank shocks affect investment? Evidence from matched bank-firm loan data, *Journal of Political Economy* 126, 525–587.
- Anderson, Michael L., 2008, Multiple inference and gender differences in the effects of early intervention: A reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects, *Journal of the American Statistical Association* 103, 1481–1495.

- Antràs, Pol, and C. Fritz Foley, 2015, Poultry in motion: A study of international trade finance practices, *Journal of Political Economy* 123, 809–852.
- Banerjee, Abhijit V., and Esther Dufo, 2014, Do firms want to borrow more? Testing credit constraints using a directed lending program, *Review of Economic Studies* 81, 572–607.
- Baxter, Marianne, and Mario J. Crucini, 1995, Business cycles and the asset structure of foreign trade, *International Economic Review* 36, 821–854.
- Becker, Thorsten, Elena Biewen, Robert Neulen, Simone Schultz, and Michael Weisbecker, 2019, Corporate Balance Sheets (Ustan) 1999-2015, Data Report 2019-01 - Metadata Version 1. Deutsche Bundesbank Research Data and Service Centre.
- Bena, Jan, Serdar Dinc, and Isil Erel, 2022, The international propagation of economic downturns through multinational companies: The real economy channel, *Journal of Financial Economics* 146, 277–304.
- Bentolila, Samuel, Marcel Jansen, and Gabriel Jiménez, 2018, When credit dries up: Job losses in the great recession, *Journal of the European Economic Association* 16, 650–695.
- Berg, Tobias, Markus Reisinger, and Daniel Streitz, 2021, Spillover effects in empirical corporate finance, *Journal of Financial Economics* 142, 1109–1127.
- Bloom, Nicholas, Raffaella Sadun, and John Van Reenen, 2012, Americans do it better: US multinationals and the productivity miracle, *American Economic Review* 102, 167–201.
- Boehm, Christoph E., Aaron Flaaen, and Nitya Pandalai-Nayar, 2019, Input linkages and the transmission of shocks: Firm-level evidence from the 2011 Tōhoku earthquake, *Review of Economics and Statistics* 101, 60–75.
- Boot, Arnoud W. A., 2000, Relationship banking: What do we know?, *Journal of Financial Intermediation* 9, 7–25.
- Boutin, Xavier, Giacinta Cestone, Chiara Fumagalli, Giovanni Pica, and Nicolas Serrano-Velarde, 2013, The deep-pocket effect of internal capital markets, *Journal of Financial Economics* 109, 122–145.
- Broner, Fernando, Tatiana Didier, Aitor Erce, and Sergio L. Schmukler, 2010, Gross capital flows: Dynamics and crises, CEPR Discussion Paper No. 8591.
- Budd, John W., Jozef Konings, and Matthew J. Slaughter, 2005, Wages and international rent sharing in multinational firms, *Review of Economics and Statistics* 87, 73–84.
- Cadestin, Charles, Koen de backer, Isabelle Desnoyers-James, Sébastien Miroudot, Davide Rigo, and Ming Ye, 2018, Multinational enterprises and global value chains: The OECD analytical AMNE database. https://www.oecd-ilibrary.org/trade/multinational-enterprises-and-global-value-chains-the-oecd-analytical-amne-database_d9de288d-en
- Calomiris, Charles W., Charles P. Himmelberg, and Paul Wachtel, 1995, Commercial paper, corporate finance, and the business cycle: A microeconomic perspective, *Carnegie-Rochester Conference Series on Public Policy* 42, 203–250.
- Carvalho, Vasco M., and Basile Grassi, 2019, Large firm dynamics and the business cycle, *American Economic Review* 109, 1375–1425.
- Catherine, Sylvain, Thomas Chaney, Zongbo Huang, David Alexandre Sraer, and David Thesmar, 2022, Quantifying reduced-form evidence on collateral constraints, *Journal of Finance* 77, 2143–2181.
- Cetorelli, Nicola, and Linda S. Goldberg, 2012, Liquidity management of U.S. global banks: Internal capital markets in the great recession, *Journal of International Economics* 88, 299–311.
- Chari, Varadarajan V., Lawrence J. Christiano, and Martin Eichenbaum, 1995, Inside money, outside money, and short-term interest rates, *Journal of Money, Credit and Banking* 27, 1354–1386.
- Chodorow-Reich, Gabriel, 2014, The employment effects of credit market disruptions: Firm-level evidence from the 2008-9 financial crisis, *Quarterly Journal of Economics* 129, 1–59.
- Contessi, Silvio, 2010, How does multinational production change international comovement? Working paper.
- Cravino, Javier, and Andrei A. Levchenko, 2017, Multinational firms and international business cycle transmission, *Quarterly Journal of Economics* 132, 921–962.
- de Haas, Ralph, and Iman van Lelyveld, 2014, Multinational banks and the global financial crisis: Weathering the perfect storm?, *Journal of Money, Credit and Banking* 46, 333–364.

- Desai, Mihir A., and C. Fritz Foley, 2006, The comovement of returns and investment within the multinational firm, in NBER International Seminar on Macroeconomics 2004, pp. 197–237.
- Desai, Mihir A., C. Fritz Foley, and Kristin J. Forbes, 2008, Financial constraints and growth: Multinational and local firm responses to currency depreciations, *Review of Financial Studies* 21, 2857–2888.
- Desai, Mihir A., C. Fritz Foley, and James R. Hines, 2004, A multinational perspective on capital structure choice and internal capital markets, *Journal of Finance* 59, 2451–2487.
- Deutsche Bundesbank, 2017a, Microdatabase Direct Investment. Version 3.0. Deutsche Bundesbank. Dataset. doi:10.12757/Bbk.MiDi.9915.03.04.
- Deutsche Bundesbank, 2017b, Corporate Balance Sheets 1999–2015. Version 2.0. Deutsche Bundesbank. Dataset. doi:10.12757/Bbk.Ustan.9915.02.02.
- di Giovanni, Julian, Şebnem Kalemli-Özcan, Mehmet Fatih Ulu, and Yusuf Soner Baskaya, 2021, International spillovers and local credit cycles, *Review of Economic Studies* 89, 733–773.
- di Giovanni, Julian, and Andrei A. Levchenko, 2012, Country size, international trade, and aggregate fluctuations in granular economies, *Journal of Political Economy* 120, 1083–1132.
- di Giovanni, Julian, Andrei A. Levchenko, and Isabelle Mejean, 2014, Firms, destinations, and aggregate fluctuations, *Econometrica* 82, 1303–1340.
- di Giovanni, Julian, Andrei A. Levchenko, and Isabelle Mejean, 2018, The micro origins of international business-cycle comovement, *American Economic Review* 108, 82–108.
- di Giovanni, Julian, Andrei A. Levchenko, and Isabelle Mejean, 2024, Foreign shocks as granular fluctuations, *Journal of Political Economy* 132, 391–433.
- Dwenger, Nadja, Frank M. Fossen, and Martin Simmler, 2015, From financial to real economic crisis: Evidence from individual firm-bank relationships in Germany, DIW Berlin Discussion Paper No. 1510.
- Eaton, Jonathan, Samuel Kortum, Brent Neiman, and John Romalis, 2016, Trade and the global recession, *American Economic Review* 106, 3401–3438.
- Elsas, Ralf, 2005, Empirical determinants of relationship lending, *Journal of Financial Intermediation* 14, 32–57.
- Farre-Mensa, Joan, and Alexander Ljungqvist, 2016, Do measures of financial constraints measure financial constraints?, *Review of Financial Studies* 29, 271–308.
- Fazzari, Steven M., and Bruce C. Petersen, 1993, Working capital and fixed investment: New evidence on financing constraints, *RAND Journal of Economics* 24, 328–342.
- Fernández, Andrés, Michael W. Klein, Alessandro Rebucci, Martin Schindler, and Martin Uribe, 2016, Capital control measures: A new dataset, *IMF Economic Review* 64, 548–574.
- Fritz Foley, C., and Kalina Manova, 2015, International trade, multinational activity, and corporate finance, *Annual Review of Economics* 7, 119–146.
- Fort, Teresa C., John Haltiwanger, Ron S. Jarmin, and Javier Miranda, 2013, How firms respond to business cycles: The role of firm age and firm size, *IMF Economic Review* 61, 520–559.
- Foster, Lucia, Cheryl Grim, and John Haltiwanger, 2016, Reallocation in the great recession: Cleansing or not?, *Journal of Labor Economics* 34, S293–S331.
- Fraccaroli, Nicolò, and Alessandro Pizzigolotto, 2022, Credit shocks and populism, Working paper.
- Frankel, Jeffrey A., and Andrew K. Rose, 1998, The endogeneity of the optimum currency area criteria, *Economic Journal* 108, 1009–1025.
- Friberg, Richard, and Mark Sanctuary, 2016, The contribution of firm-level shocks to aggregate fluctuations: The case of Sweden, *Economics Letters* 147, 8–11.
- Fuchsbriefe, 2018, Unternehmen verzichten auf billige Kredite. <https://www.fuchsbriefe.de/betrieb/finanzierung/unternehmen-verzichten-auf-billige-kredite>
- Gabaix, Xavier, 2011, The granular origins of aggregate fluctuations, *Econometrica* 79, 733–772.
- Giroud, Xavier, Simone Lenzu, Quinn Maingi, and Holger M. Mueller, 2021, Propagation and amplification of local productivity spillovers, NBER Working Paper No. 29084.
- Giroud, Xavier, and Holger M. Mueller, 2015, Capital and labor reallocation within firms, *Journal of Finance* 70, 1767–1804.
- Giroud, Xavier, and Holger M. Mueller, 2017, Firm leverage, consumer demand, and employment losses during the great recession, *Quarterly Journal of Economics* 132, 271–316.

- Giroud, Xavier, and Holger M. Mueller, 2019, Firms' internal networks and local economic shocks, *American Economic Review* 109, 3617–3649.
- Gopalan, Radhakrishnan, Vikram Nanda, and Amit Seru, 2007, Affiliated firms and financial support: Evidence from Indian business groups, *Journal of Financial Economics* 86, 759–795.
- Graham, John R., and Krishnamoorthy Narasimhan, 2004, Corporate survival and managerial experiences during the great depression, Working paper.
- Grassi, Basile, 2017, IO in I-O: Size, industrial organization, and the input-output network make a firm structurally important, Working paper.
- Guadalupe, Maria, Olga Kuzmina, and Catherine Thomas, 2012, Innovation and foreign ownership, *American Economic Review* 102, 3594–3627.
- Gumpert, Anna, Kalina Manova, Cristina Rujan, and Monika Schnitzer, 2023, Multinational firms and global innovation.
- Hassan, Tarek A., Stephan Hollander, Laurence van Lent, and Ahmed Tahoun, 2024, The global impact of Brexit uncertainty, *Journal of Finance* 79, 413–458.
- Hassan, Tarek A., Jesse Schreger, Markus Schwedeler, and Ahmed Tahoun, 2024, Sources and transmission of country risk, *Review of Economic Studies* (forthcoming).
- Herreño, Juan, 2020, The aggregate effects of bank lending cuts, Working paper.
- Huber, Kilian, 2018, Disentangling the effects of a banking crisis: Evidence from German firms and counties, *American Economic Review* 108, 868–898.
- Imbs, Jean, 2004, Trade, finance, specialization, and synchronization, *Review of Economics and Statistics* 86, 723–734.
- Imbs, Jean, 2006, The real effects of financial integration, *Journal of International Economics* 68, 296–324.
- Inderst, Roman, and Holger M. Mueller, 2003, Internal versus external financing: An optimal contracting approach, *Journal of Finance* 58, 1033–1062.
- Kalemli-Özcan, Şebnem, Herman Kamil, and Carolina Villegas-Sanchez, 2016, What hinders investment in the aftermath of financial crises: Insolvent firms or illiquid banks?, *Review of Economics and Statistics* 98, 756–769.
- Kalemli-Özcan, Şebnem, Elias Papaioannou, and José-Luis Peydro, 2013, Financial regulation, financial globalization, and the synchronization of economic activity, *Journal of Finance* 68, 1179–1228.
- Koetter, Michael, Felix Noth, and Oliver Rehbein, 2020, Borrowers under water! Rare disasters, regional banks, and recovery lending, *Journal of Financial Intermediation* 43, 100811.
- Krueger, Miriam, Cordula Munzert, and Harald Stahl, 2017, External Position of Banks 03/2002-12/2016, Data Report 2017-05 - Metadata Version 1, Deutsche Bundesbank Research Data and Service Centre (RDSC).
- Lamont, Owen, 1997, Cash flow and investment: Evidence from internal capital markets, *Journal of Finance* 52, 83–109.
- Lemmon, Michael, and Michael R. Roberts, 2010, The response of corporate financing and investment to changes in the supply of credit, *Journal of Financial and Quantitative Analysis* 45, 555–587.
- Lian, Chen, and Yueran Ma, 2020, Anatomy of corporate borrowing constraints, *Quarterly Journal of Economics* 136, 229–291.
- Magerman, Glenn, Karolien De Bruyne, Emmanuel Dhyne, and Jan Van Hove, 2017, Heterogeneous firms and the micro origins of aggregate fluctuations, Working paper.
- Malmendier, Ulrike, Geoffrey Tate, and Jon Yan, 2011, Overconfidence and early-life experiences: The effect of managerial traits on corporate financial policies, *Journal of Finance* 66, 1687–1733.
- Manova, Kalina, Shang-Jin Wei, and Zhiwei Zhang, 2015, Firm exports and multinational activity under credit constraints, *Review of Economics and Statistics* 97, 574–588.
- Matvos, Gregor, and Amit Seru, 2014, Resource allocation within firms and financial market dislocation: Evidence from diversified conglomerates, *Review of Financial Studies* 27, 1143–1189.
- Menno, Dominik, 2017, Multinational firms and business cycle transmission, Working paper.
- Mezzanotti, Filippo, and Timothy Simcoe, 2022, Research and/or development? Financial frictions and innovation investment, Working paper.

- Monnet, Eric, and Damien Puy, 2019, One ring to rule them all? New evidence on world cycles, IMF WP/19/202.
- Morgan, Donald P., Bertrand Rime, and Philip E. Strahan, 2004, Bank integration and state business cycles, *Quarterly Journal of Economics* 119, 1555–1584.
- Ongena, Steven, José-Luis Peydró, and Neeltje van Horen, 2015, Shocks abroad, pain at home? Bank-firm-level evidence on the international transmission of financial shocks, *IMF Economic Review* 63, 698–750.
- Overesch, Michael, 2006, Transfer pricing of intrafirm sales as a profit shifting channel—Evidence from German firm data, ZEW Discussion Paper No. 06-084, Mannheim.
- Paravisini, Daniel, Veronica Rappoport, Philipp Schnabl, and Daniel Wolfenzon, 2015, Dissecting the effect of credit supply on trade: Evidence from matched credit-export data, *Review of Economic Studies* 82, 333–359.
- Peek, Joe, and Eric S. Rosengren, 1997, The international transmission of financial shocks: The case of Japan, *American Economic Review* 87, 495–505.
- Popov, Alexander, and Gregory Udell, 2012, Cross-border banking, credit access, and the financial crisis, *Journal of International Economics* 87, 147–161.
- Puri, Manju, Jörg Rocholl, and Sascha Steffen, 2011, Global retail lending in the aftermath of the US financial crisis: Distinguishing between supply and demand effects, *Journal of Financial Economics* 100, 556–578.
- Rajan, Raghuram, Henri Servaes, and Luigi Zingales, 2000, The cost of diversity: The diversification discount and inefficient investment, *Journal of Finance* 55, 35–80.
- Ramondo, Natalia, Veronica Rappoport, and Kim J. Ruhl, 2016, Intrafirm trade and vertical fragmentation in U.S. multinational corporations, *Journal of International Economics* 98, 51–59.
- Reuters, 2008, Continental asks banks to reschedule debt.
- Reuters, 2018, VW extends \$21 billion loan to weather Diesel fall-out.
- Santioni, Raffaele, Fabio Schiantarelli, and Philip E. Strahan, 2020, Internal capital markets in times of crisis: The benefit of group affiliation, *Review of Finance* 24, 773–811.
- Scharfstein, David S., and Jeremy C. Stein, 2000, The dark side of internal capital markets: Divisional rent-seeking and inefficient investment, *Journal of Finance* 55, 2537–2564.
- Schild, Christopher-Johannes, Simone Schultz, and Franco Wieser, 2017, Linking Deutsche Bundesbank Company Data Using Machine-Learning-Based Classification, Technical Report 2017-01, Deutsche Bundesbank Research Data and Service Centre.
- Schild, Christopher-Johannes, and Frank Walter, 2017, Microdatabase Direct Investment 1999–2015, Data Report 2017-01 - Metadata Version 4, Deutsche Bundesbank Research Data and Service Centre.
- Schnabl, Philipp, 2012, The international transmission of bank liquidity shocks: Evidence from an emerging market, *Journal of Finance* 67, 897–932.
- Seru, Amit, 2014, Firm boundaries matter: Evidence from conglomerates and R&D activity, *Journal of Financial Economics* 111, 381–405.
- Setzler, Bradley, and Felix Tintelnot, 2021, The effects of foreign multinationals on workers and firms in the United States, *Quarterly Journal of Economics* 136, 1943–1991.
- Sharpe, Steven A., 1990, Asymmetric information, bank lending and implicit contracts: A stylized model of customer relationships, *Journal of Finance* 45, 1069–1087.
- Shin, Hyun-Han, and René M. Stulz, 1998, Are internal capital markets efficient?, *Quarterly Journal of Economics* 113, 531–552.
- Slovin, Myron B., Marie E. Sushka, and John A. Polonchek, 1993, The value of bank durability: Borrowers as bank stakeholders, *Journal of Finance* 48, 247–266.
- Sraer, David, and David Thesmar, 2023, How to use natural experiments to estimate misallocation, *American Economic Review* 113, 906–938.
- Stein, Jeremy C., 1997, Internal capital markets and the competition for corporate resources, *Journal of Finance* 52, 111–133.
- Tintelnot, Felix, 2017, Global production with export platforms, *Quarterly Journal of Economics* 132, 157–209.
- Zlate, Andrei, 2016, Offshore production and business cycle dynamics with heterogeneous firms, *Journal of International Economics* 100, 34–49.

Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.